

The TMS9918 The First Sprite Chip

By Karl M. Gutttag

KGOnTech Blog (www.kgutttag.com)

My TI Timeline (Dates Roughly Reconstructed)

- BSEE Bradley '76 MSEE Michigan '77
- 1977-07 Joined TI Houston/Stafford
 - Ford EEC-III at TI cancelled day I joined
- 1977-08 Worked on floppy disc controller (TMS9909)
 - Pulled off and the 9909 program was transfer to TI-UK
- 1977-09 to 1998-06 Designed and Debugged TMS9918
- 1978-07 Helped debug TMS9985
 - Learned about the 9900 and 9985 architecture
- 1979-01 Lead architect of TMS9995
 - Spent 6 months in Bedford UK (designers in UK)
- 1979-10 Lead architect of TMS99000 (99105 and 99110)
 - 3-micron (skipped 4-micron) NMOS (based on 9995 design)
 - In US working with some former 9918 team members
- 1982-01 Started defining TMS34010 and VRAM (TMS4161)
 - Defined jointly in US and UK with software support in Houston
 - Designed in TI Bedford UK (In UK from March to Sept. 1983)
- 1982 Elected Senior Member of Technical Staff
 - Youngest in the history of TI
- 1983 Help define TMS34061 (VRAM controller) and TMS 34074 (4-bit Color Palette (RAMDAC))
- 1985 Started TMS34020
 - Started discussion with Apple for Color Mac
- 1986-09 TMS34010 Shipped
 - "Battled" with Intel 82786 and others
 - Dominated "high end" graphics processor market 1987-1990
- 1988 Elected TI Fellow (Youngest in the history of TI)
- 1988-09 TMS34020 Announced
- 1989 Help with definition of first SDRAM
 - As a result of my work with DRAMs, Graphics, and CPUs
- 1989-1995 Defined and Design TMS320C80 Image Processor
 - Defined in Houston, Designed in UK
 - Announced March 1994
- 1996 Started on Successor to TMS320C80
- 1997 Left TI to join Silicon Display

Since Leaving TI

- 1998-1999 Silicon Display at CTO
 - Founded by Jerry Rogers (9918 & founder of Cyrix)
 - Head worn display using LCOS
- 2000-2003 Developed new Digital LCOS Backplane
 - Programmable SIMD-like counting on backplane
- 2004-2011 Founded and CTO of Syndiant (LCOS)
 - First product 3-Chip targeting rear projection television
 - Pivoted to field sequential color aimed a pico-projectors both stand alone and cell phone
 - Started getting into head worn displays
- **2011-Present Started KGOntech Blog**
 - Initially to discuss work at TI
 - Turned into a blog on microdisplay devices and later focused on augmented reality display and optics
 - 10K to 30K readers per month
- 2013-2015 CTO of Navdy
 - Aftermarket automotive HUD
 - Develop the overall design and optics using TI-DLP display engine
- 2018-2022 RAVN Chief Scientist
 - Developing lightweight military AR glasses
 - Company shifted to more pure software

TMS9918 When I started (September 1977)

- DRAM Based (Designed for 4K Bit 4027 DRAM)
 - **First consumer chip to use DRAMs**
 - The theory was that by the time the 9918 came out 4K DRAMs would be very inexpensive, but DRAM companies stopped making 4K DRAMs – led to 9918A to support 16K DRAMs
- Originally five Sprites (“Sprites” named by David Ackley)
 - Sprite “pre-processing” came later (by Karl and Pete)
- 5 Micron Ratioless NMOS – TI used more heavily than other companies
 - Lower power than static NMOS but trickier to design
- Name mapping to pattern (common low-cost video technique)
- 5.4xxx MHz ($= 3.58\text{xxx color burst} \times 3 \div 2$)
 - Do or die speed (no speed binning)
 - BTW – the first IBM PC used 4.77 MHz ($= \text{color burst} \times 4 \div 3$)
- CPU Interface was not defined
- 40-pin plastic package (for cost reasons)

Dave Ackley's "Sprites"

Quoting Dave on what he did on sprites:

Dynamic multilayer graphics - Movable Sprites one in front of another and in front of the MxN 8x8 or 16x16 with 15 colors plus transparent graphic patterns for a background and the Sprites.

*The key invention of the Sprites was that they could be moved quickly and easily by simply changing their starting X and Y register values, plus these Sprite images were passed **thru a priority encoder** which with the transparent color allowed Sprites to appear to be in front of each other but at the same time to be able to selectively see through parts of a sprite. As an example, one could display a person riding a bike in front of some stores and at the same time see these stores through the spokes of the bicycle's tires.*

This hardware invention allowed creating rapid smooth motion/animation which was otherwise impossible using the slow processors of the day."

Dave was working with Milton Bradley and Granville Ott (designer of the TI 99/4) during the early definition of the 9918. This was all before I was assigned to the 9918.

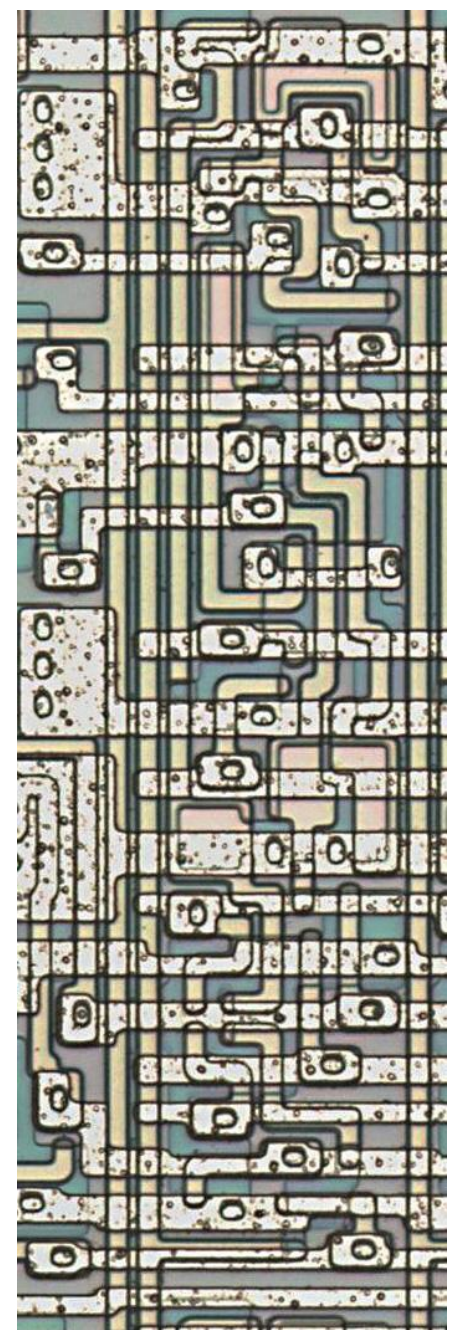
TMS 9918 Design Team

- Karl Gutttag
 - Sprite definition, logic, & circuit design
 - DRAM interface definition
 - Whole device logic simulation/verification
 - Briefly worked on a microcoded 9918 alternative
- Pete Macourek
 - Sprite Definition
 - CPU definition and design
 - Whole device functional simulation/verify
- Ki Suk (Richard) Chang
 - Design manager
 - Clock and I/O Buffer Design
 - Any difficult circuit design
- Segio Maggi
 - Vertical and Horizontal Logic and Circuit Design
- Joe Sexton
 - Analog, Clock and Buffer Design
 - Video interface circuitry
- Chris ???
 - Logic and circuit design
 - Temporary assignment to learn MOS
- Larry Monks, Denise Blalock, Joe Peters, & Claudia Kyle
 - I.C. layout (hand drawn transistors on mylar)
- Jerry Rogers (later founded Cyrix)
 - Program Manager
- David Ackley
 - Created the name “Sprites” & much of how they worked with priority encoder (before pre-processing)
 - Early 9918 Definition – Dave was promoted to a new position soon after Karl and Pete started on the 9918

TMS9918 Technology (1977)

- 5-micron self aligned polysilicon gate N-MOS
- Single metal layer, single polysilicon layer, and depletion
- 5V (some internal bootstrap nodes “gated clocks”)
- 5.4xx Mhz clock rate
- Dynamic/Ratioless N-MOS
 - Very few static logic gates
- 3" (76.2mm) wafer
- ~35K-40K total transistors
- 40 pin plastic (sometimes ceramic) package

One bit of
Sprite horizontal
Down counter



Ratioless NMOS

Pros

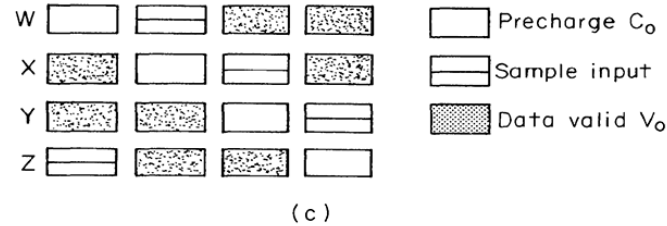
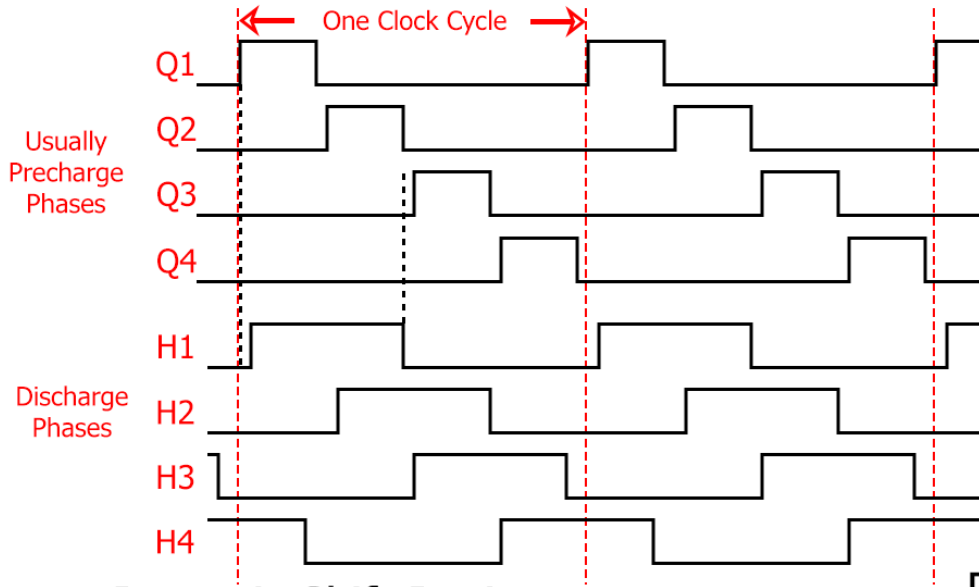
- Lower power than static NMOS
- Smaller transistor sizes
- Support for complex functions
 - Particularly good for ALUs and Counters
 - “Domino logic” everywhere
- TI had experience in Ratioless
 - Used in PMOS calculator chips

Cons

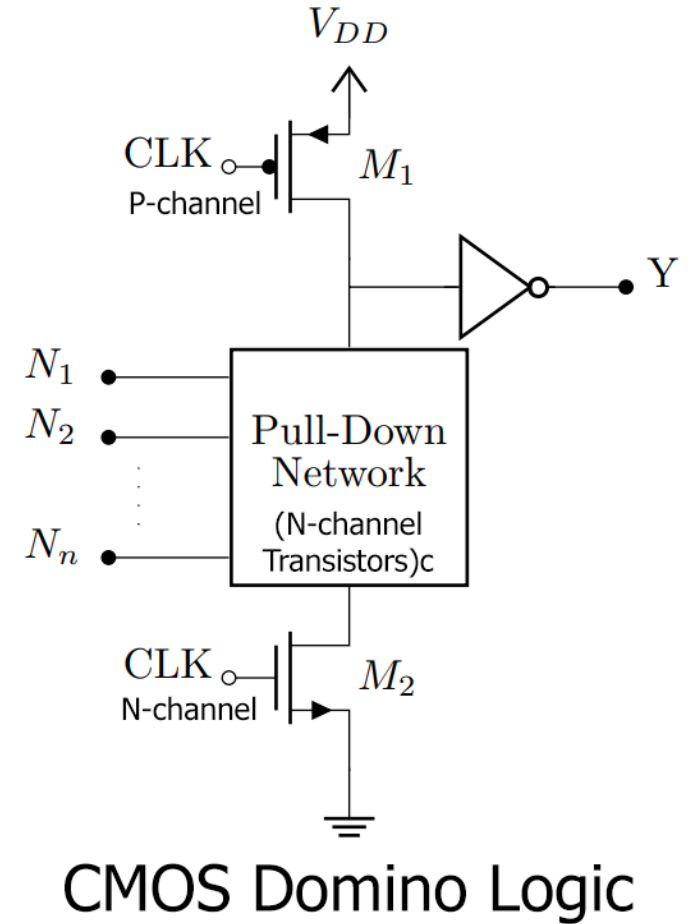
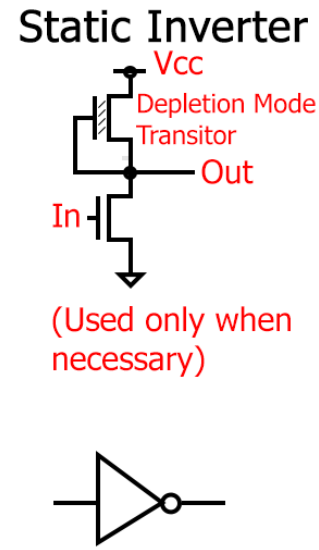
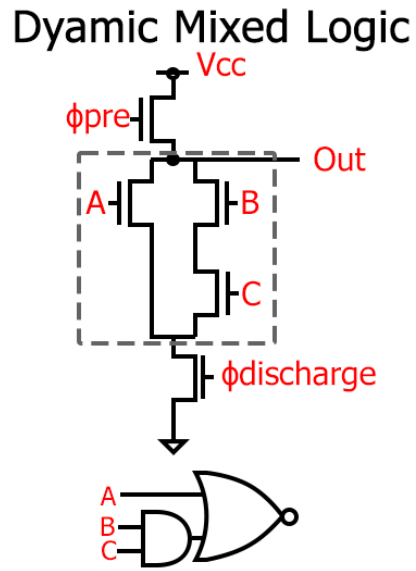
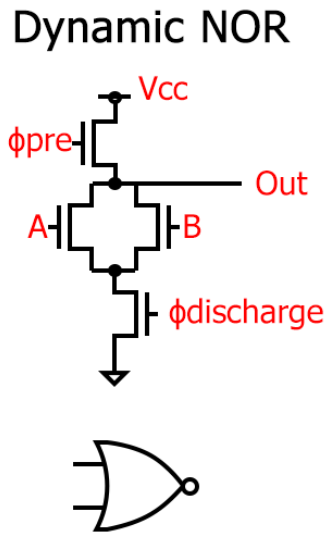
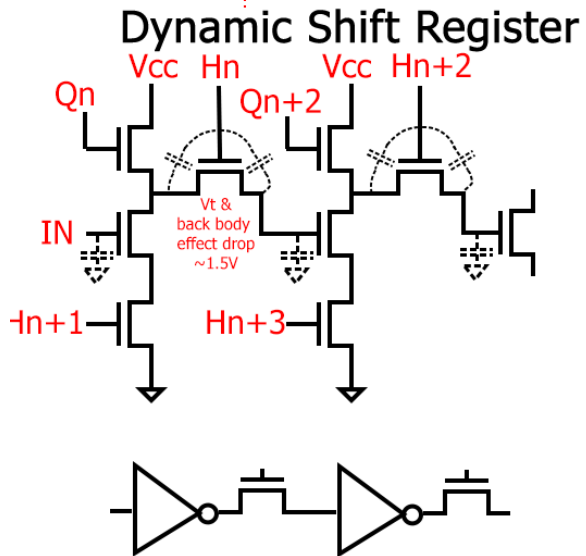
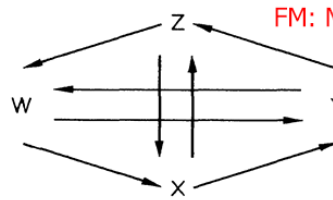
- Trickier to design
- Requires huge clock drivers and busses
 - Clock power (still a win over static NMOS)
- Gated clocks (Bootstrapped)
- **Floating nodes everywhere**
 - Can’t stop clock or even run very slow
 - Harder to test and verify
 - More likely to have intermittent problems
- TI was so proficient at Ratioless design that it may have help lull it in to delaying the transition to CMOS

Ratioless NMOS

Four (4) phase set of clocks



FM: MOS/LSI Design and Application
1972 by CARR and Mize
(available on Bitsavers)



9918 Designed "Little Endian"

3.7.2 TMS9928A/9929A Terminal Assignments

SIGNATURE	TERMINAL	I/O	DESCRIPTION
XTAL1, XTAL2	40,39	I	10.7 + MHz crystal inputs*
R-Y	38	O	VDP color burst frequency clock. On the TMS9928A/9929A, this is the R-Y color difference output.
GROMCLK	37	O	VDP output clock = XTAL/24. Typically not used.
Y	36	O	Composite video output. On the TMS9928A/9929A, this is the Y (black/white luminance and composite sync) output.
B-Y	35	I/O	External VDP input. On the TMS9928A/9929A, this is the B-Y color difference output.
RESET/ SYNC	34	I	The RESET pin is a trilevel input pin. When it is below 0.8 volts, RESET initializes the VDP. When it is above 9 volts, RESET is the synchronizing input for external video.
VCC	33	I	+ 5 volt supply
RD0 MSB	32	I	VRAM read data bus
RD1	31	I	
RD2	30	I	
RD3	29	I	
RD4	28	I	
RD5	27	I	
RD6	26	I	
RD7	25	I	
CD0 MSB	24	I/O	CPU data bus; (CD0) is the most significant bit
CD1	23	I/O	
CD2	22	I/O	
CD3	21	I/O	
CD4	20	I/O	
CD5	19	I/O	
CD6	18	I/O	
CD7 LSB	17	I/O	
INT	16	O	CPU interrupt output.
CSR	15	I	CPU-VDP read strobe
CSW	14	I	CPU-VDP write strobe
MODE	13	I	CPU interface mode select; usually a processor address line

RAS	1	40	XTAL2
CAS	2	39	XTAL1
AD7	3	38	R-Y
AD6	4	37	GROMCLK
AD5	5	36	Y
AD4	6	35	B-Y
AD3	7	34	RESET/SYNC
AD2	8	33	VCC
AD1	9	32	RD0
AD0	10	31	RD1
R/W	11	30	RD2
VSS	12	29	RD3
MODE	13	28	RD4
CSW	14	27	RD5
CSR	15	26	RD6
INT	16	25	RD7
CD7	17	24	CD0
CD6	18	23	CD1
CD5	19	22	CD2
CD4	20	21	CD3

SIGNATURE	TERMINAL	I/O	DESCRIPTION
VSS	12	I	Ground References
R/W	11	O	VRAM write strobe
AD0 MSB	10	O	VRAM address/data bus (multiplexed high and low order VRAM address and output data bytes)
AD1	9	O	AD0 is the most significant bit and is used only for data and not for addressing.**
AD2	8	O	Caused by the 9918 being designed "little endian" (LSB=0) and then cosmetically converted to big endian
AD3	7	O	
AD4	6	O	
AD5	5	O	
AD6	4	O	
AD7	3	O	
CAS	2	O	VRAM row address strobe
RAS	1	O	VRam row address strobe

* When driven externally, both inputs must be driven.

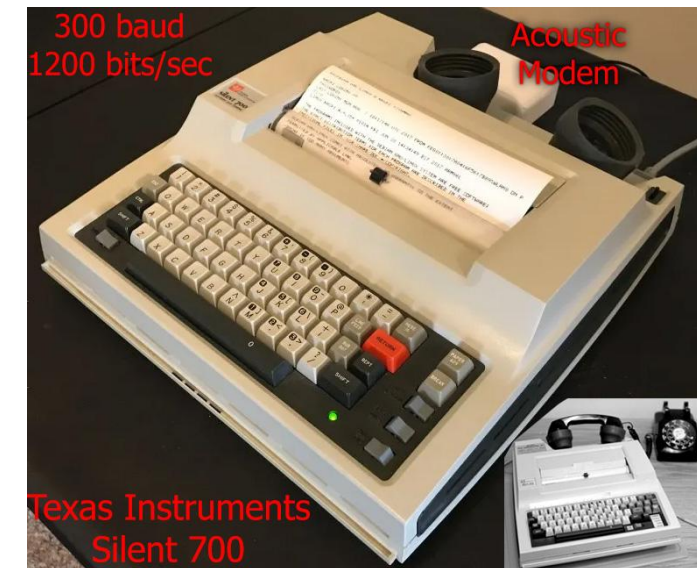
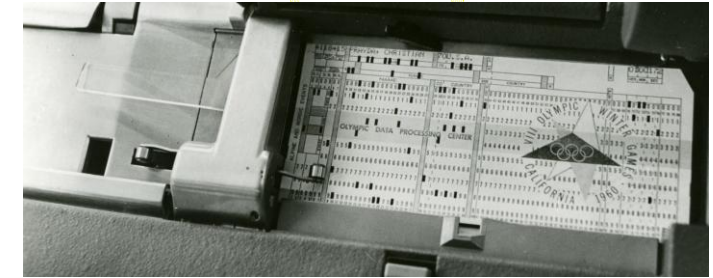
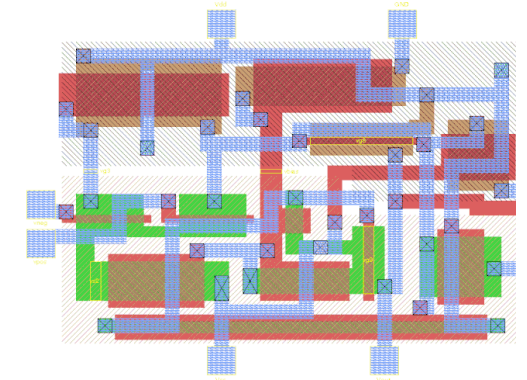
** The least significant address bit (AD7) is wired to A0 of the dynamic RAMs. Likewise, AD6 is wired to A1 of the RAMs.

* When driven externally, both inputs must be driven.

** The least significant address bit (AD7) is wired to A0 of the dynamic RAMs. Likewise, AD6 is wired to A1 of the RAMs.

How the 9918 was designed

- Logic with paper and pencil
- Transistors/circuitry drawn with colored pencils on mylar plastic
 - Hand digitized
 - Large 4-pen flat bed printer printed composite for hand/eye verification
- ESIM (TI internal) logic verification program (IBM 370 mainframe)
 - Monochrome IBM terminal for input (could crash between saves)
 - Silent 700 for editing and running from home
 - Primarily simulated at most one line at a time (initialized to critical point)
 - I estimated that all the logic simulation runs combined would equal about one full screen
- Spice used for circuit design – on IBM card decks (run on IBM 370)
 - I thought this was primitive when I arrived (I had used DECwriter at Michigan)
- Everything in the design was done “little endian” (LSB=0)
 - Cosmetically changed to big endian by marketing (causing problems particularly with address line connections)

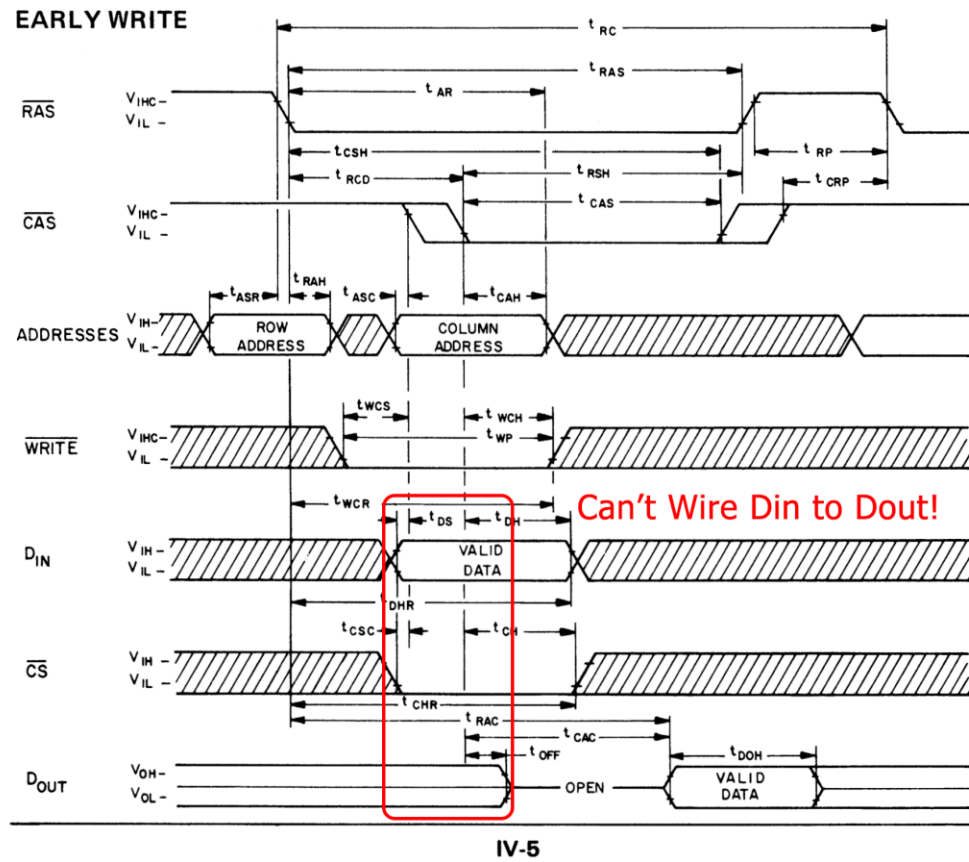


Asynchronous DRAM

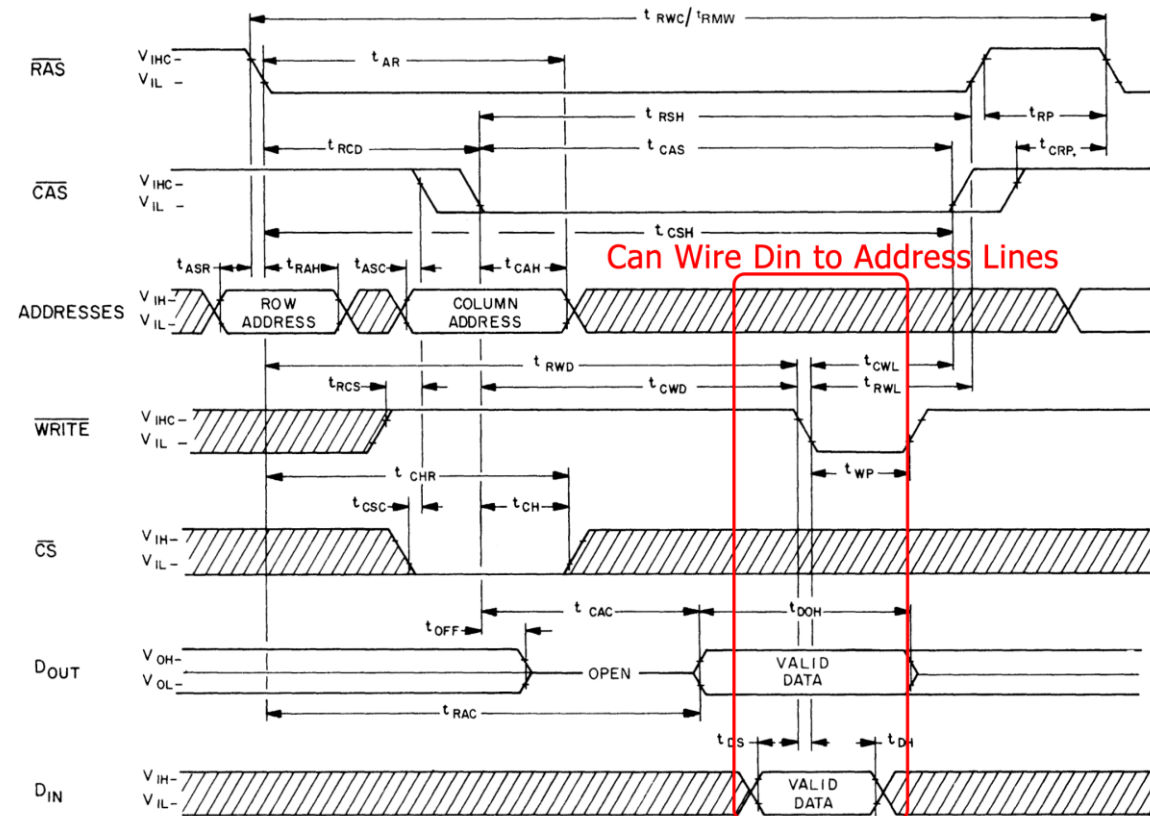
- DRAM have **pages of interlocking** timing spec's
 - The “topline” cycle time can only be achieved with precise edges
 - Mainframe computers used precision analog delay lines
 - We only had so many edges and had to allow process variations
- The 9918 was designed to the Mostek 4027-15 (4K by 1) spec
 - TI's 4K by 1 was supposed to have an identical spec
- By-1 DRAMs had separate Dout and Din pins
 - Wiring them together would cause drive conflicts between them
 - We wanted to save the cost of an external data in consumer product
- Bi-Directions buffers are bigger than Input or Output buffers
- Page-mode only gave only a 50% speed improvement (Unlike later SDRAM)
 - Indirect (“Name”) addressing interleaved with sprite pre-processing and CPU accesses resulted in no sequential accesses
 - Too complex and too much buffering (and too little benefit) to support on the 9918

By-1 DRAM Didn't Support Wiring Din and Dout Pins Together

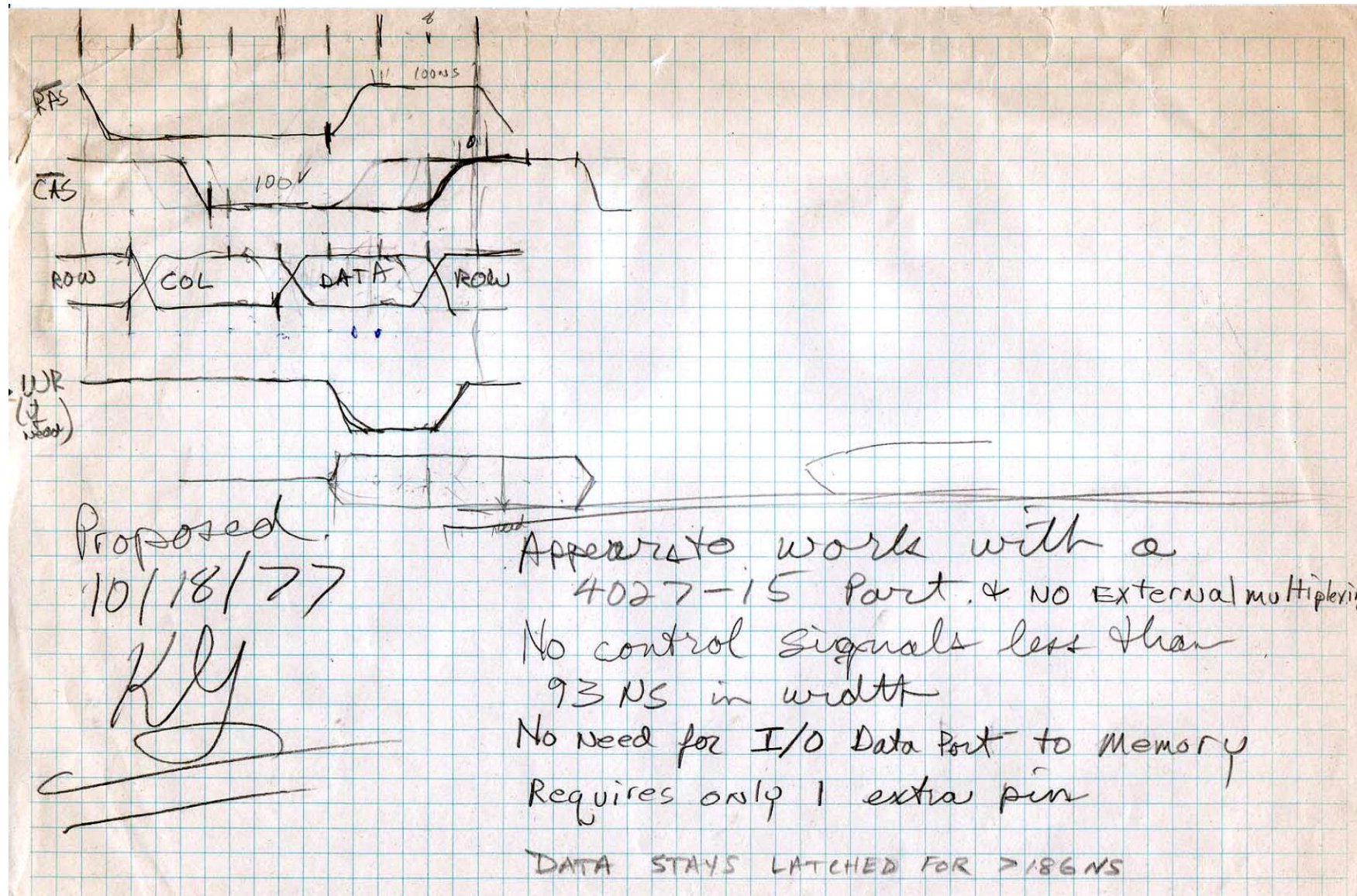
Mostek 4027 (4K by 1) DRAM Early Write



READ-WRITE / READ-MODIFY-WRITE CYCLE Also known as "Late Write"



My Proposal to Write Conflict Problem



TMS9918 DRAM Write and Read Cycles

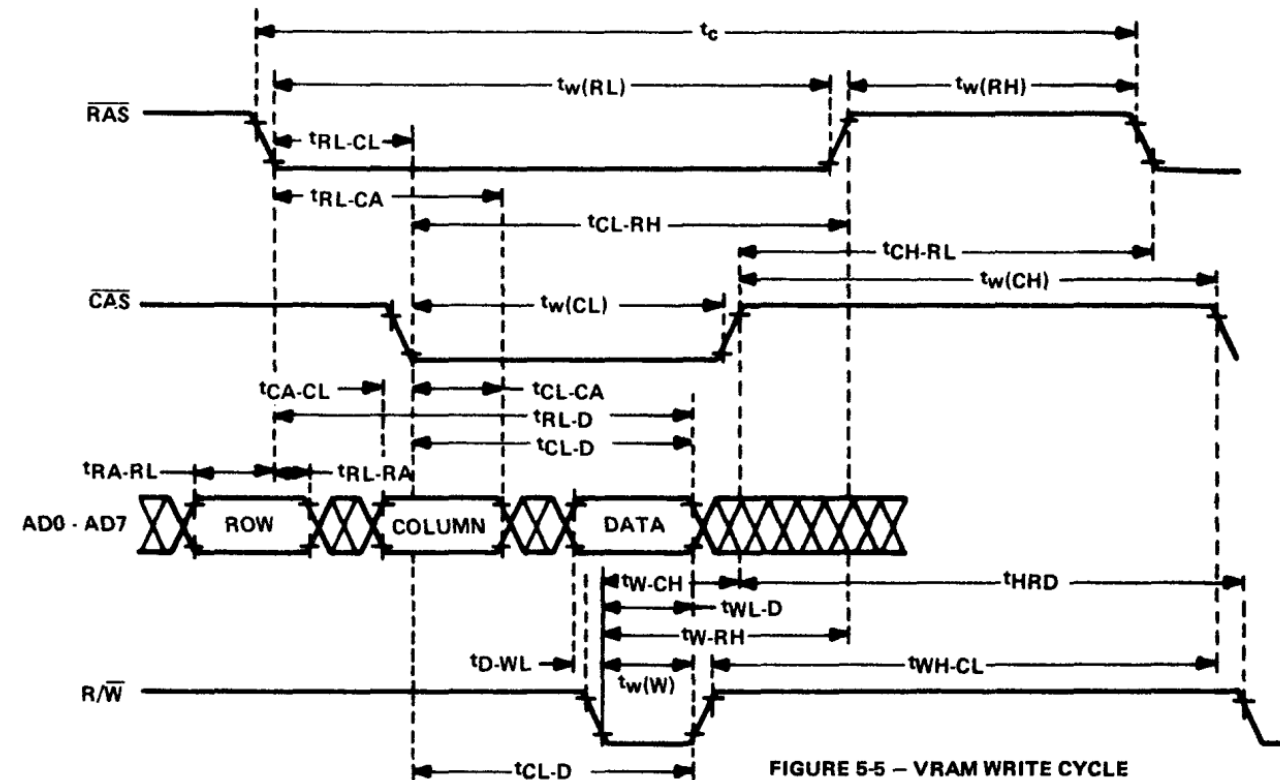


FIGURE 5-5 – VRAM WRITE CYCLE

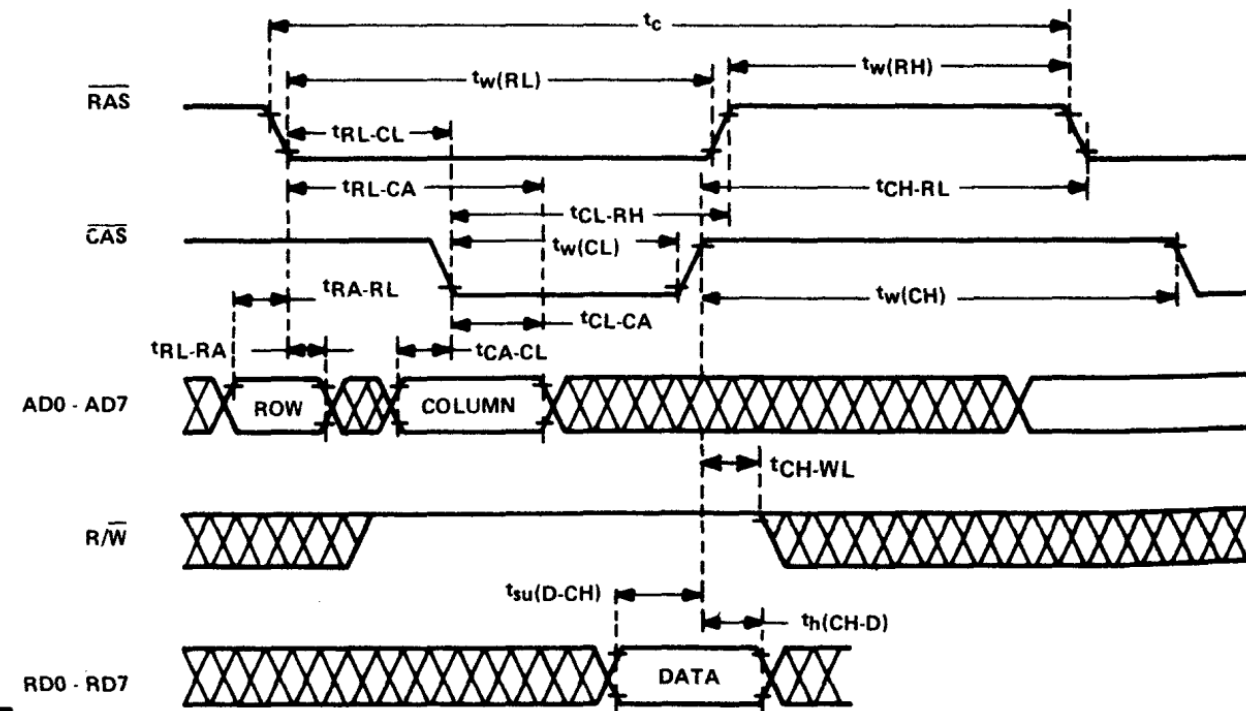


FIGURE 5-6 – VRAM READ CYCLE

NOTE: All measurements are made at 10% and 90% points.

CRT Display and DRAM Bandwidth Drove Everything

- Two Clock/Pixel Cycles per DRAM access (~ 372 nS / ~ 5.4 MHz)
 - Speed locked to Color Burst ($\sim 3.58 \times 3 / 2$)
 - Two clocks
 - Analog DRAMs require many edges giving us 8 clock edges (two, four phase clocks)
 - DRAM's 320ns "spec" not possible with only 8 clock edges (two 9918 cycles)
 - A few edges used analog delays (had to allow for process variation)
- 128 Memory cycles during the 256-pixel active screen
- 43 Memory cycles during horizontal blanking

TMS9918 CPU Interface

- Pete Macourek defined and interacted with TI's Home Computer Group
 - Pete fought against using the terrible TI CRU serial interface
 - Memory pointer/counter interface
 - Defined and designed "little endian" (LSB=0) – cosmetically changed in documents after the design was complete
- We/I didn't understand at the time that the HCG was going to use the 9918's VRAM as main memory

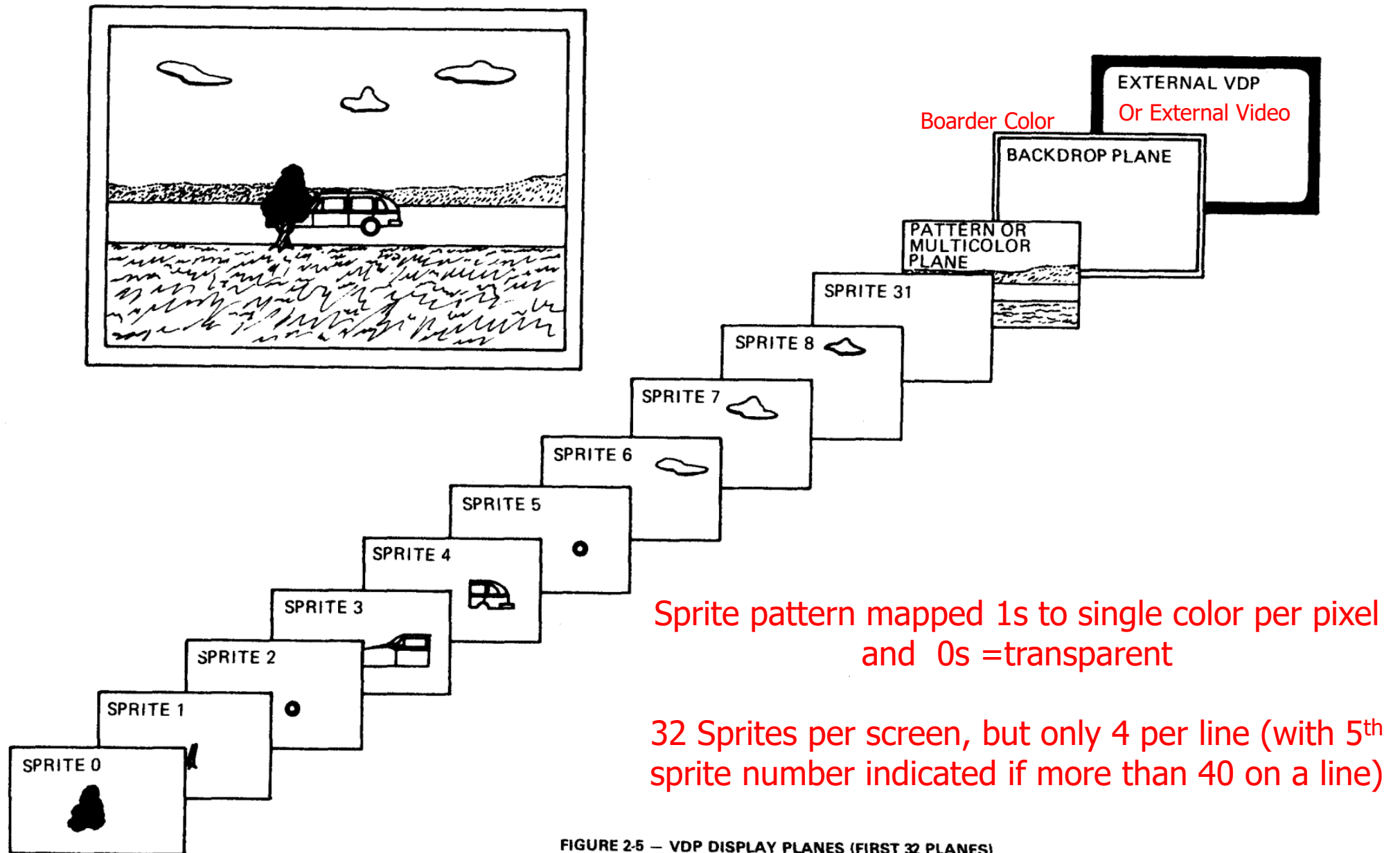


FIGURE 2-5 — VDP DISPLAY PLANES (FIRST 32 PLANES)

On chip registers

- Primarily base pointers into memory

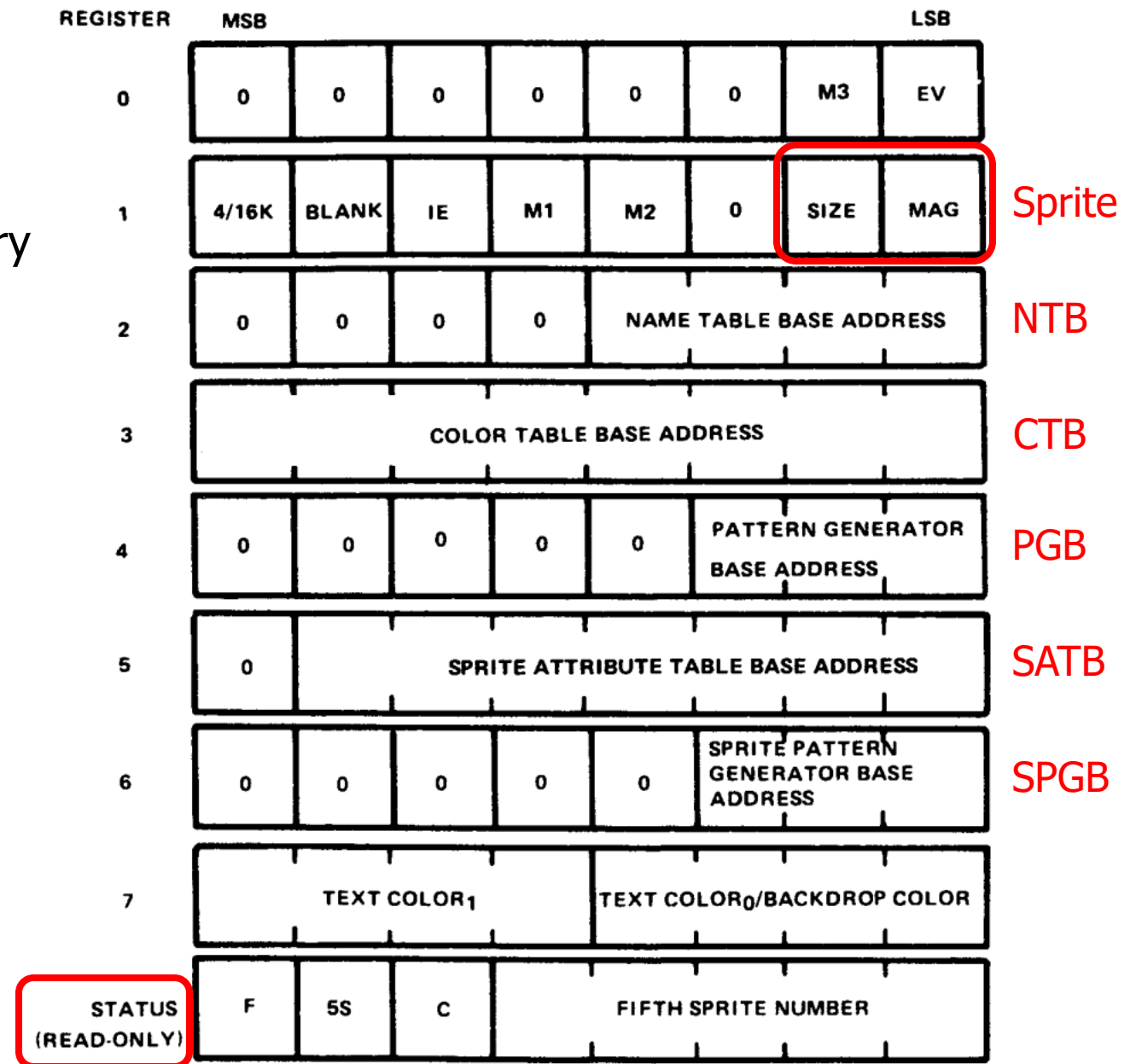
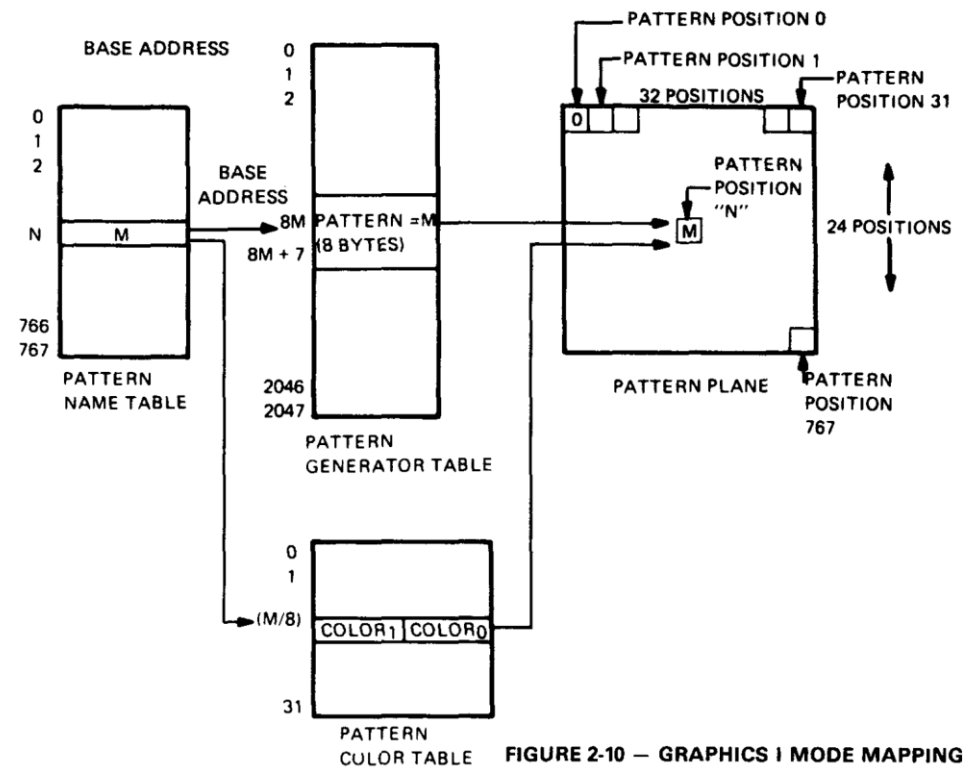


FIGURE 2-3 — VDP REGISTERS

Pattern Graphics Mapping

- Neither the storage nor bandwidth to support 4-bit bitmap (256x192x4=49K pixels / 25K bytes)
- Pattern method common (basis of text displays)
 - Form of indirect addressing to reuse shapes
 - Color mapping of binary pattern
- Pattern Base addresses (NTB, COLB, PGB) in register bits on VDP
- Pattern **N**ame 1st Memory Fetch
 - Address = NTB, Row (Vert/8), Column (Horz/8)
- Pattern **C**olors 2nd memory fetch
 - Address = COLB, 0, Name (5-MS Bits)
- Pattern (binary) is 3rd memory fetch
 - Address = PGB, **N**ame



GRAPHICS I MODE ADDRESS LOCATION																
ADDRESS TYPE		0	1	2	3	4	5	6	7	8	9	10	11	12	13	COMMENTS
1)	PATTERN NAME ADDRESS	NTB														PATTERN NAME TABLE BASE (VDP REG2) PATTERN POSITION
						ROW										
								COLUMN								
2)	PATTERN COLOR ADDRESS	COLB														PATTERN COLOR TABLE BASE (VDP REG3) ALWAYS "0" IN BIT 8 FIVE MOST SIGNIFICANT BITS OF NAME
								0								
								NAME (0-4)								
3)	PATTERN GENERATOR ADDRESS	PGB												PATTERN GENERATOR BASE (VDP REG4) ALL 8 BITS OF NAME THREE LSB'S FORM PATTERN ROW POSITION		
					NAME											
												XXX				

Multicolor Mode and Graphics 2 Mode Mapping

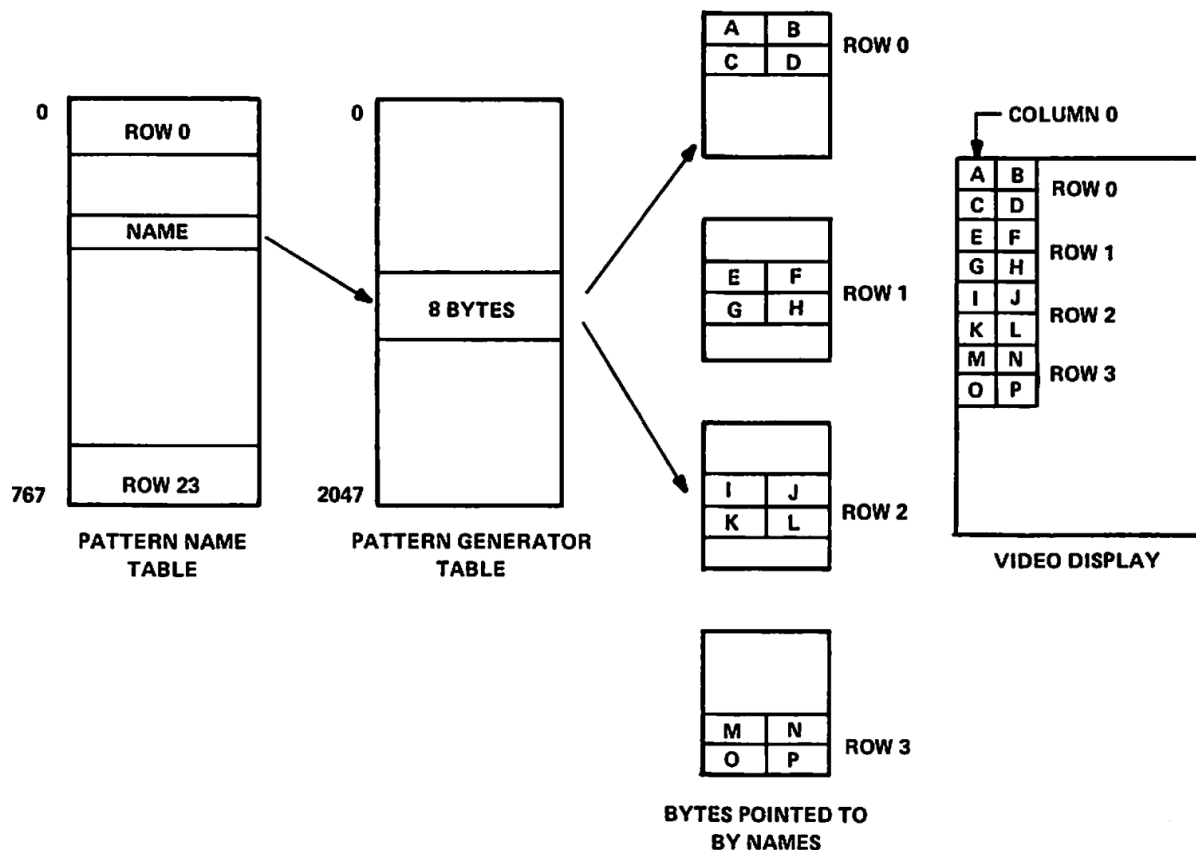


FIGURE 2-14 — MULTICOLOR MODE MAPPING

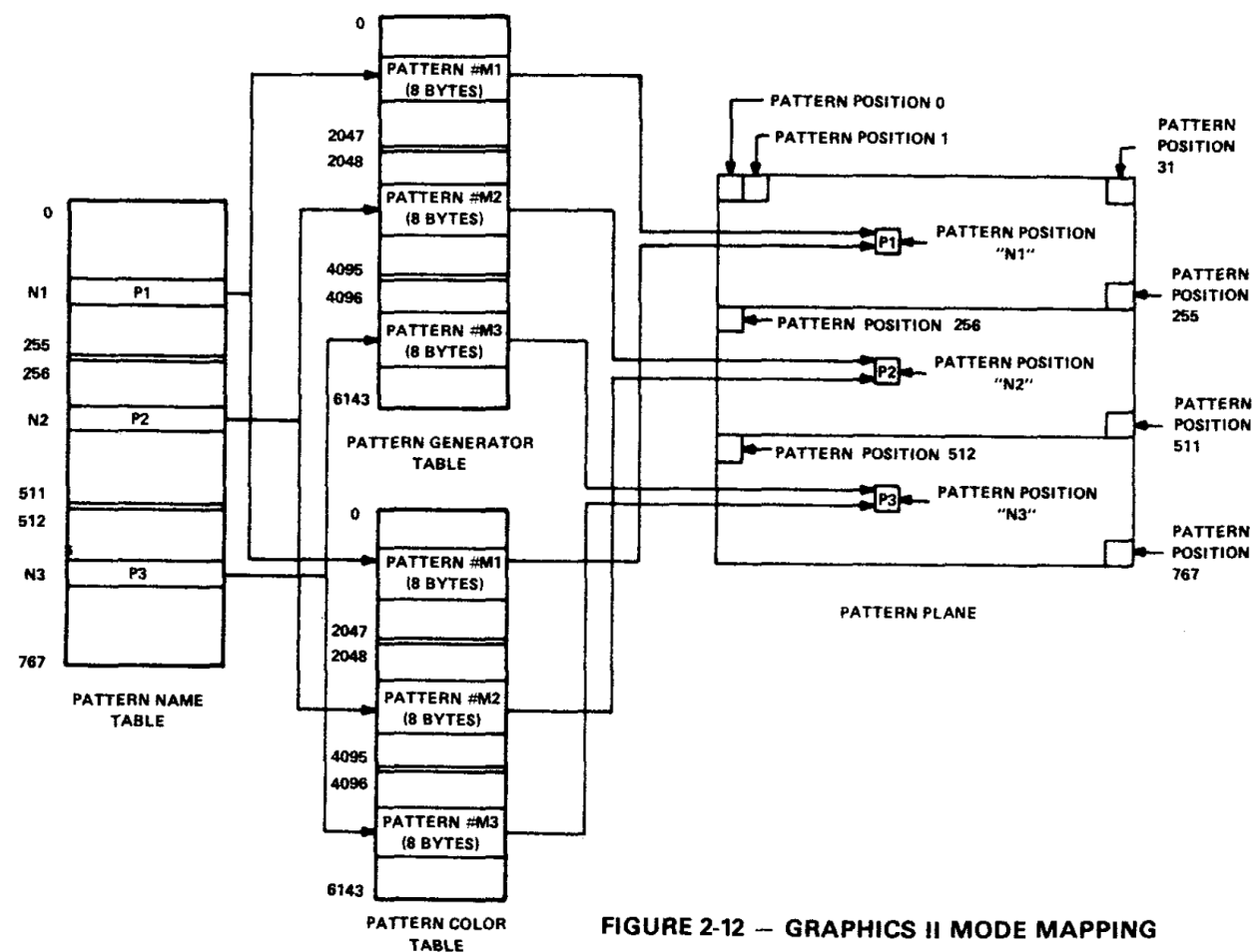


FIGURE 2-12 — GRAPHICS II MODE MAPPING

Sprite Addresses Mapping

- “Active Sprite” a function of vertical position, size and mag
- Must fetch **V**ertical, **H**orizontal, **N**ame (pointer), **A**tttribute
 - Attribute = Color, & Early Clock Bit (~9th bit of Horizontal)
- Line segment of sprite pattern pointed to by SPGB, Name, current vertical line number (one or two fetches based on pattern size)

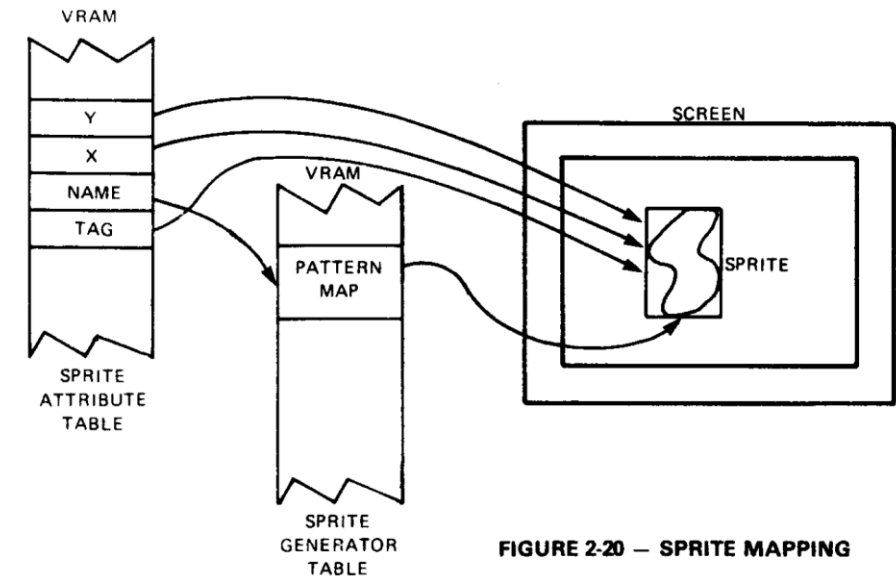


FIGURE 2-20 — SPRITE MAPPING

Sprite Attribute Table

BYTE	BIT							
	0	1	2	3	4	5	6	7
0	VERTICAL POSITION							
1	HORIZONTAL POSITION							
2	NAME							
3	EARLY CLOCK BIT	0	0	0	COLOR CODE			

FIGURE 2-19 — SPRITE ATTRIBUTE TABLE ENTRY

SIZE	MAG	AREA	RESOLUTION	BYTES/PATTERN
0	0	8 × 8	single pixel	8
1	0	16 × 16	single pixel	32
0	1	16 × 16	2 × 2 pixels	8
1	1	32 × 32	2 × 2 pixels	32

TABLE 2-5 — SPRITE PATTERN FORMATS

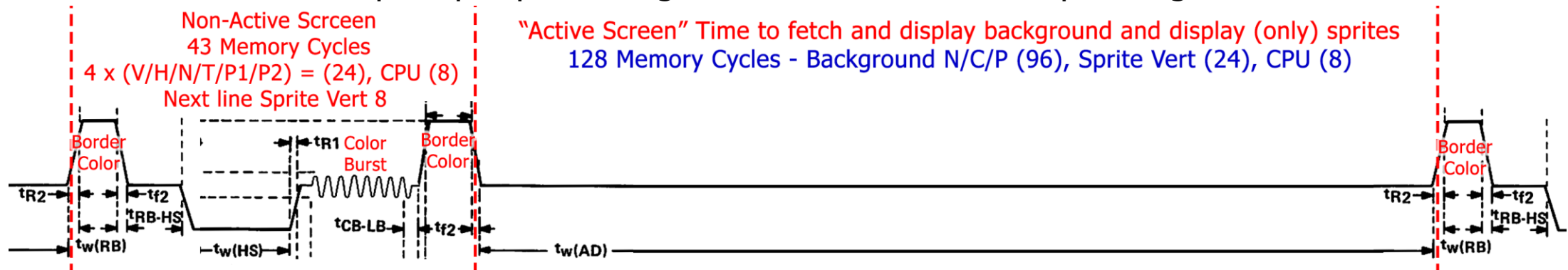
TABLE 3-2 — PATTERN GRAPHICS ADDRESS LOCATION TABLES (CONTINUED)

SPRITE ADDRESS LOCATION															
ADDRESS TYPE	0	1	2	3	4	5	6	7	8	9	10	11	12	13	COMMENTS
SPRITE ATTRIBUTE ADDRESS	SAB						SPRITE				XX				SPRITE ATTRIBUTE TABLE BASE (VDP REG5) SPRITE NUMBER ATTRIBUTE NUMBER: 00 FOR VERTICAL POSITION 01 FOR HORIZONTAL POSITION 10 FOR NAME 11 FOR TAG (EARLY CLOCK AND COLOR)
SIZE = 0 SPRITE PATTERN GENERATOR	SPGB		NAME						XXX					SPRITE PATTERN GENERATOR BASE (VDP REG4) NAME ATTRIBUTE OF SPRITE THREE LSB'S GIVE BYTE/LINE NUMBER	
SIZE = 1 SPRITE PATTERN GENERATOR	SPGB		NAME (0-5)						XXXXX						

Size and mag were “global” for all sprites to avoid memory accesses

Where Sprite Pre-Processing Came From

- 128 Memory cycles during the 256-pixel active screen
 - CPU required 8 cycles - Left 24 cycle but we could not afford to double buffer sprites
- 43 Memory cycles during horizontal blanking
 - Required reserve at least 5 for CPU access – net ~38 cycles
 - Each Sprite required at least 6 cycles (for 16-pixel wide sprites) memory cycles
 - Vertical, Horizontal, Name, Tag, plus two for pattern
- Pete and I gave to management the following options
 - 5 Sprites X 6 access/sprite = 30 accesses
 - 32 Sprites per screen, but 4 per line using “pre-processing”
 - Less Die Area – Sprite pre-processing smaller than one set of sprite registers & counters



Sprite Pre-Processing ~ Processing

- Hardware and control took over 1/3rd of whole device

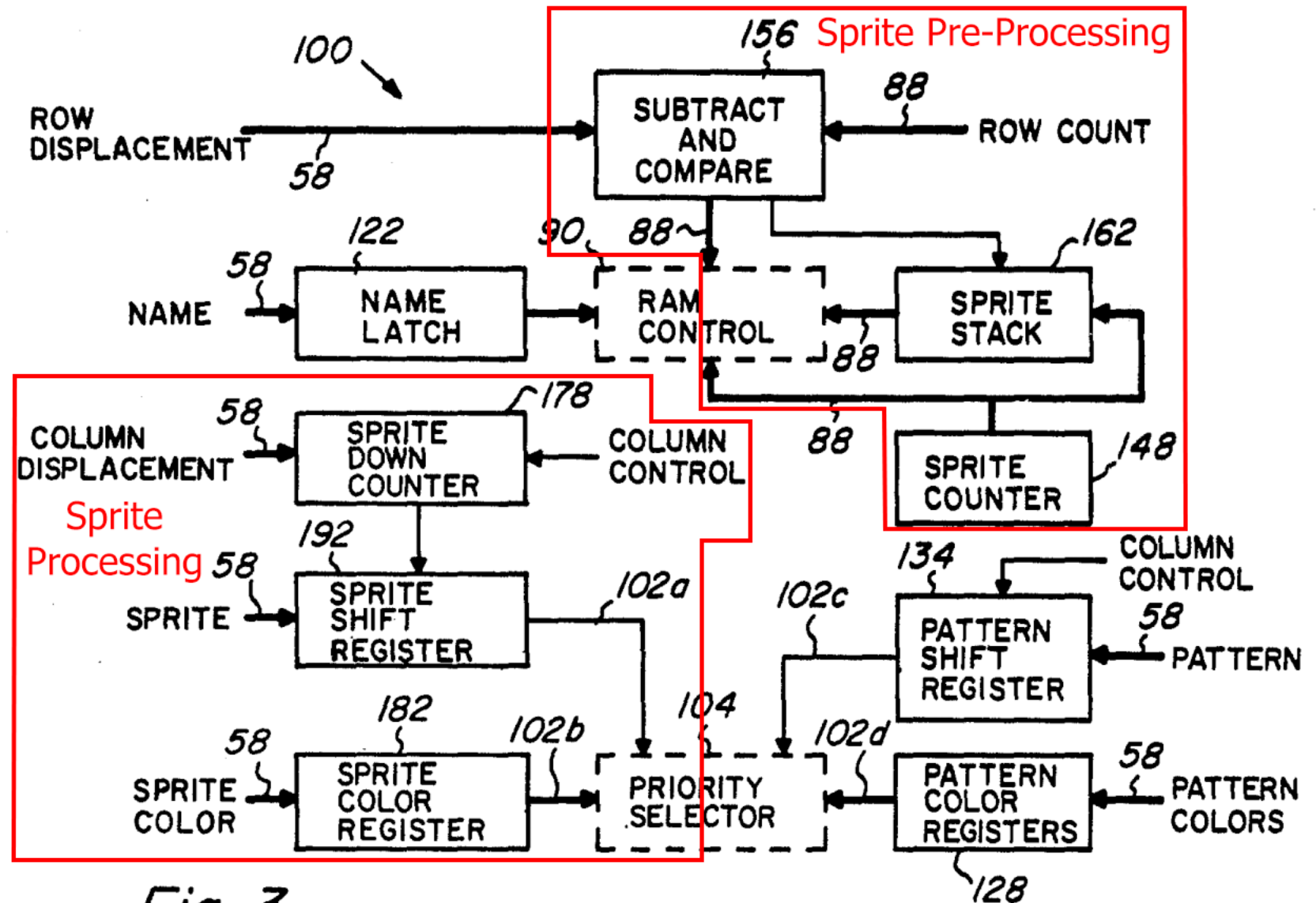
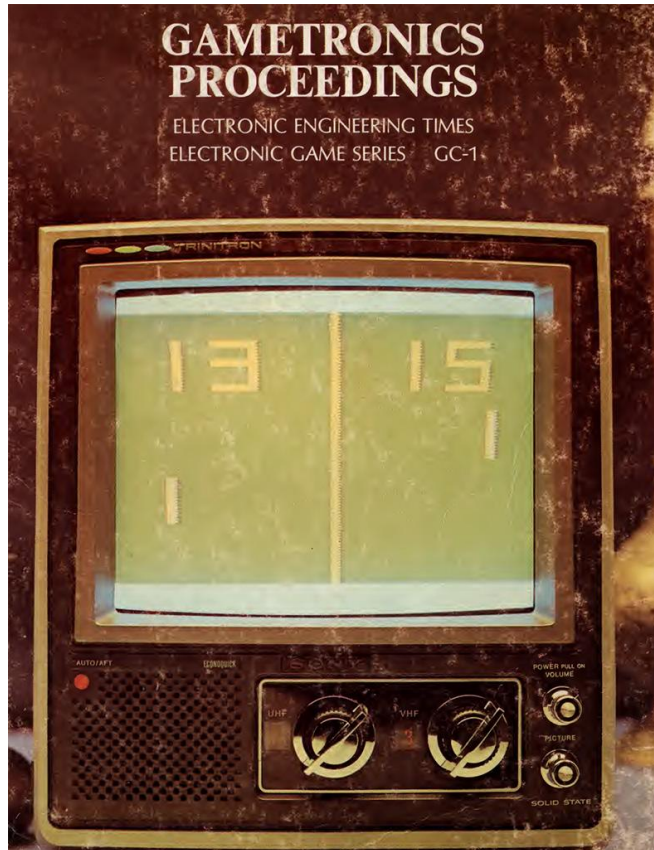


Fig. 3

How Sprites Differed From Movable/Player Objects

- A fixed number of player objects means that the hardware is dedicated to each object
- 9918 Sprites involved three “softer” processes
 1. Pre-processing which checks to see which sprites are going to be shown on a given line and saves those sprite numbers on a stack
 2. Processing (done during horizontal blanking)
 - a) Pop sprite number from stack and use it to fetch the vertical, horizontal, name (pointer to shape), and Attribute (color)
 - b) Using the name and vertical count, fetch the pattern (one or two bytes)
 - c) For each sprite, save horizontal position (for down counting), pattern, and color. Note: this hardware is not dedicated to any given sprite but is used based on the up to four sprites that can see on a line
 3. Display (during the active display) for each visible sprite
 - a) Down count until zero and start shift register shifting out pattern
 - b) Use the bits shifted out as control to a priority selector between the sprite and pattern colors

"Player/Object Graphics" before "Sprites"



I was given these proceedings when I started on the 9918

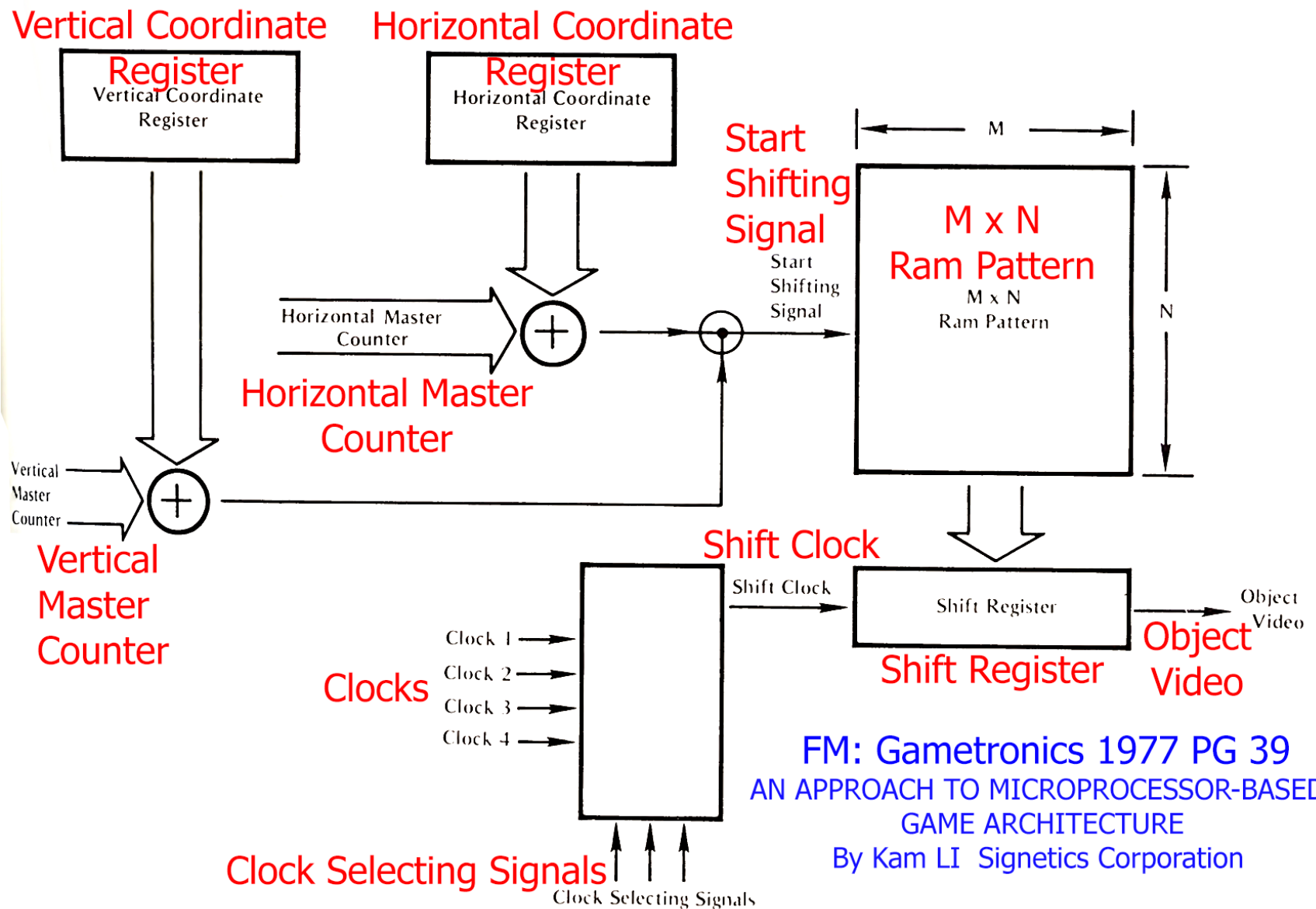
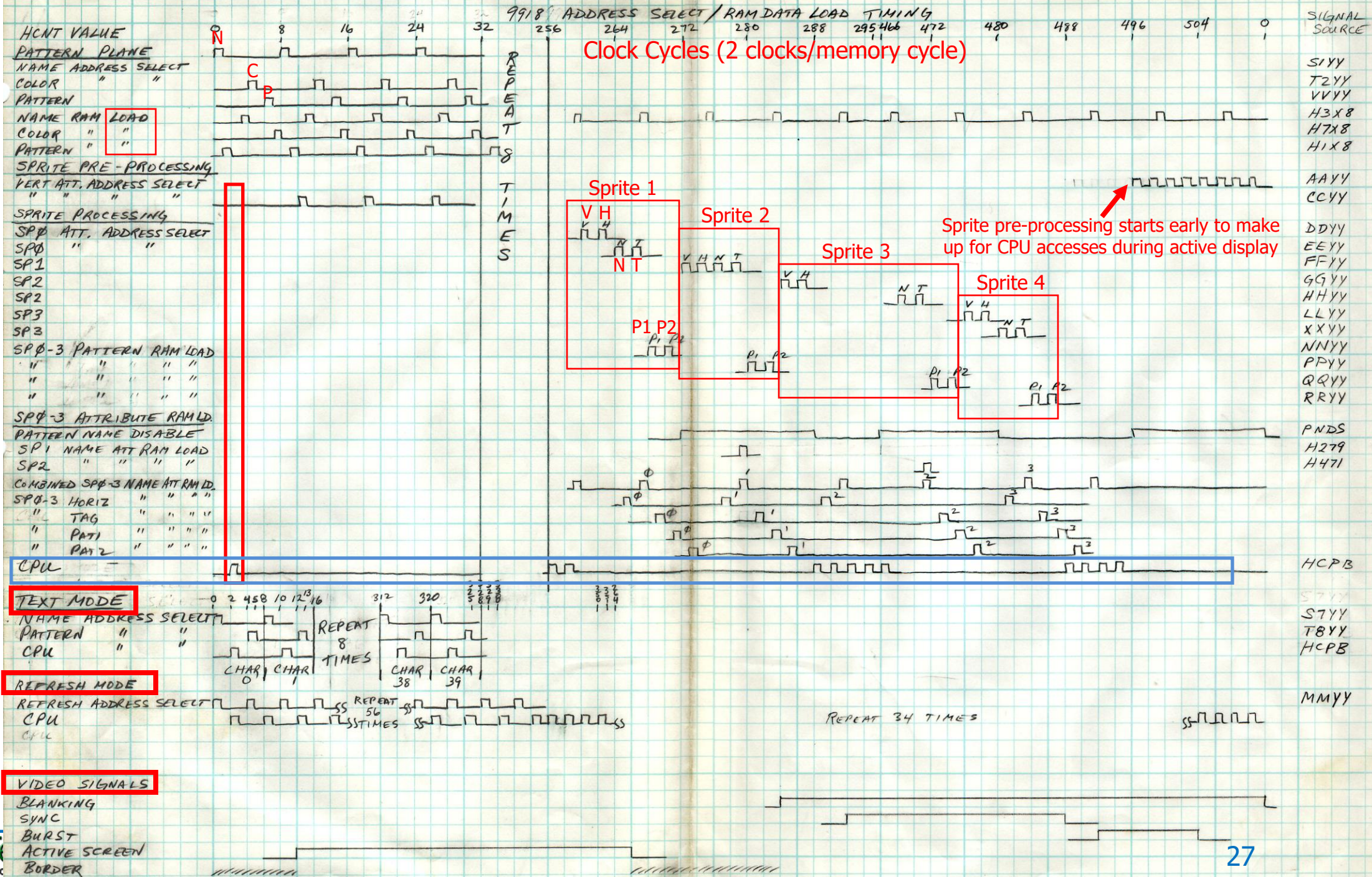


Figure 5 Object Module Block Diagram

Figure 5 Object Module Block Diagram

FM: Gametronics 1977 PG 39
AN APPROACH TO MICROPROCESSOR-BASED
GAME ARCHITECTURE
By Kam LI Signetics Corporation

Vertical Timing Diagram by Sergio Maggi



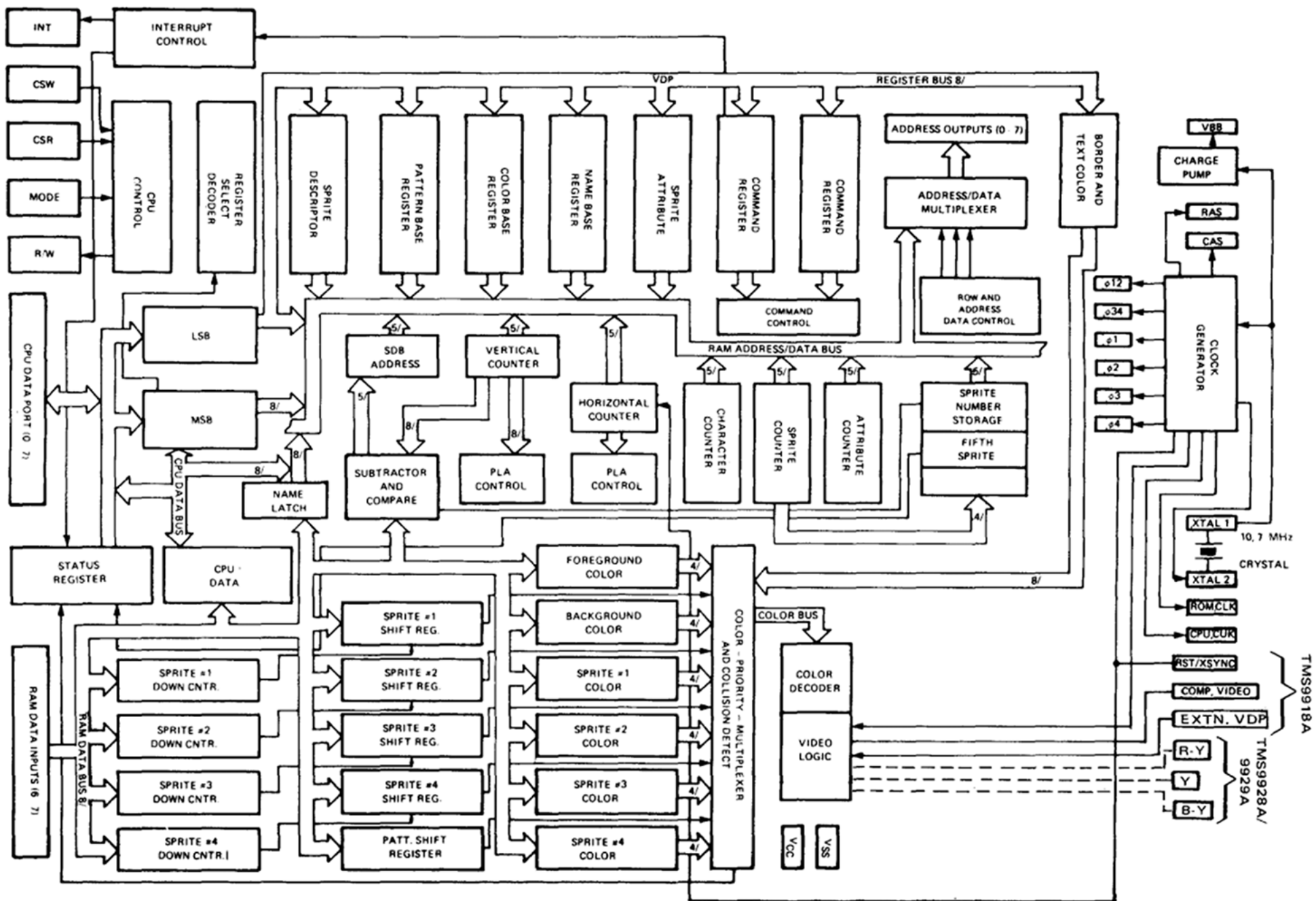
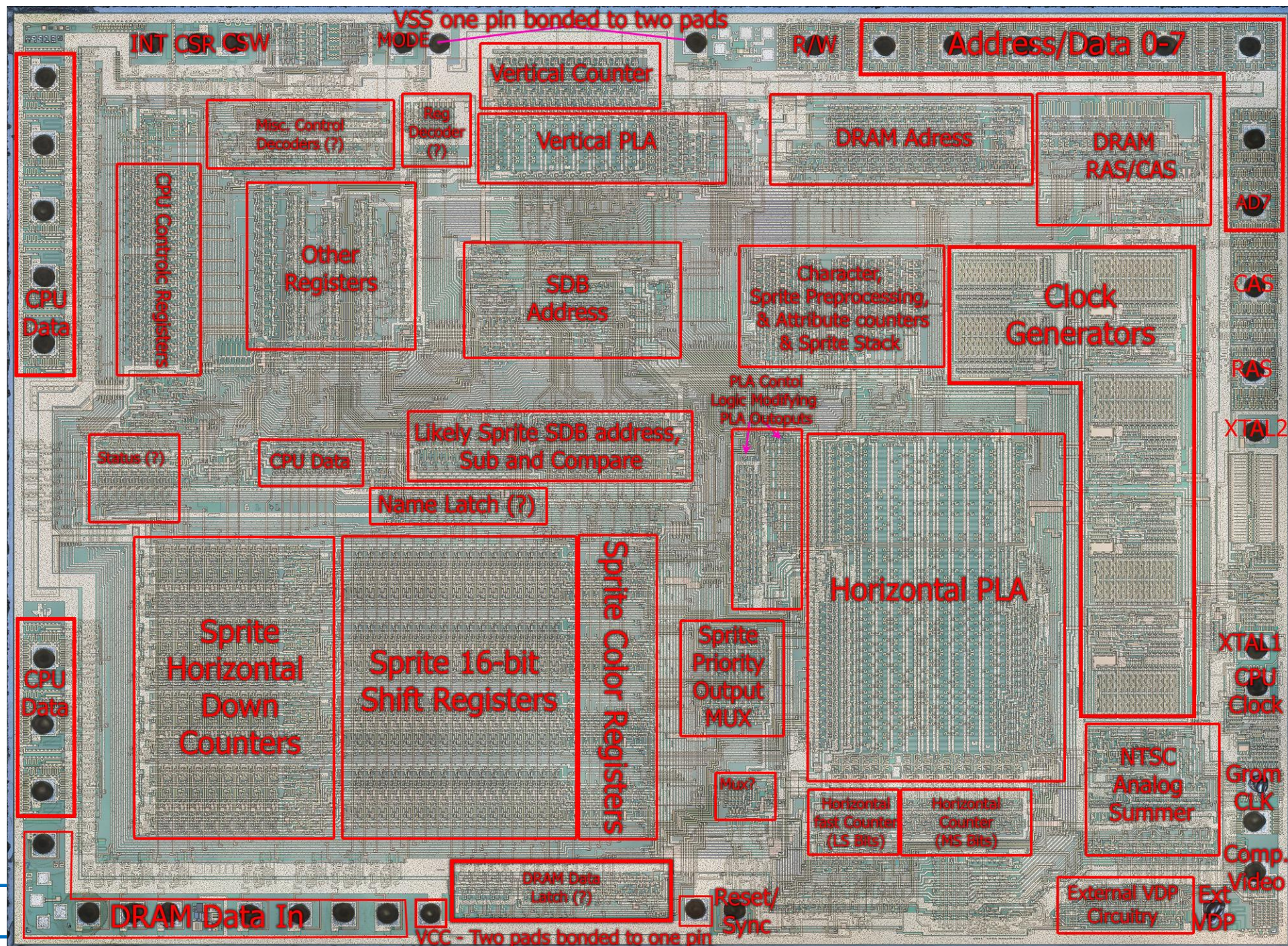
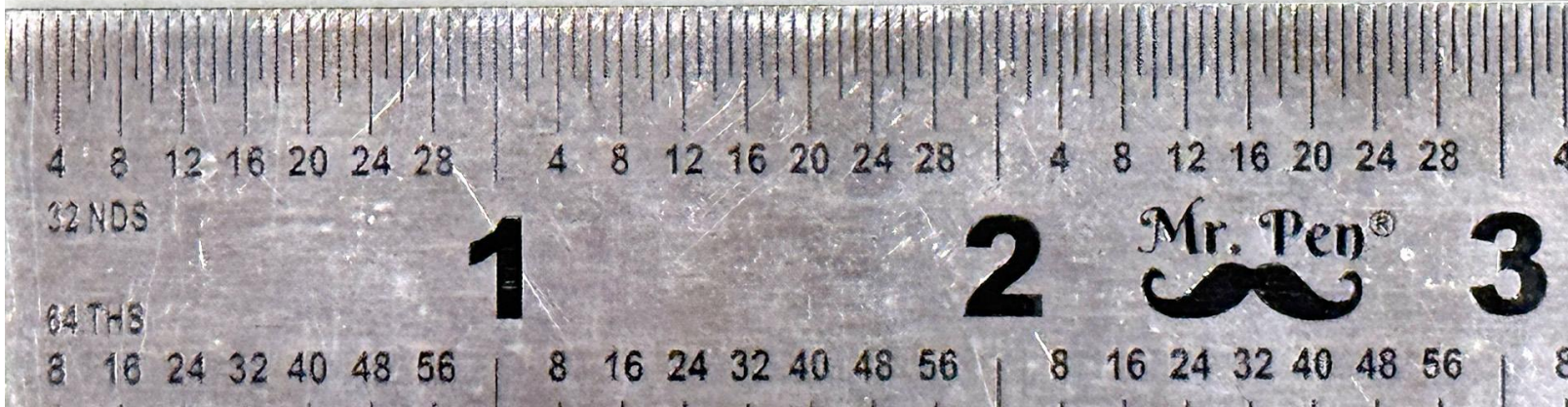
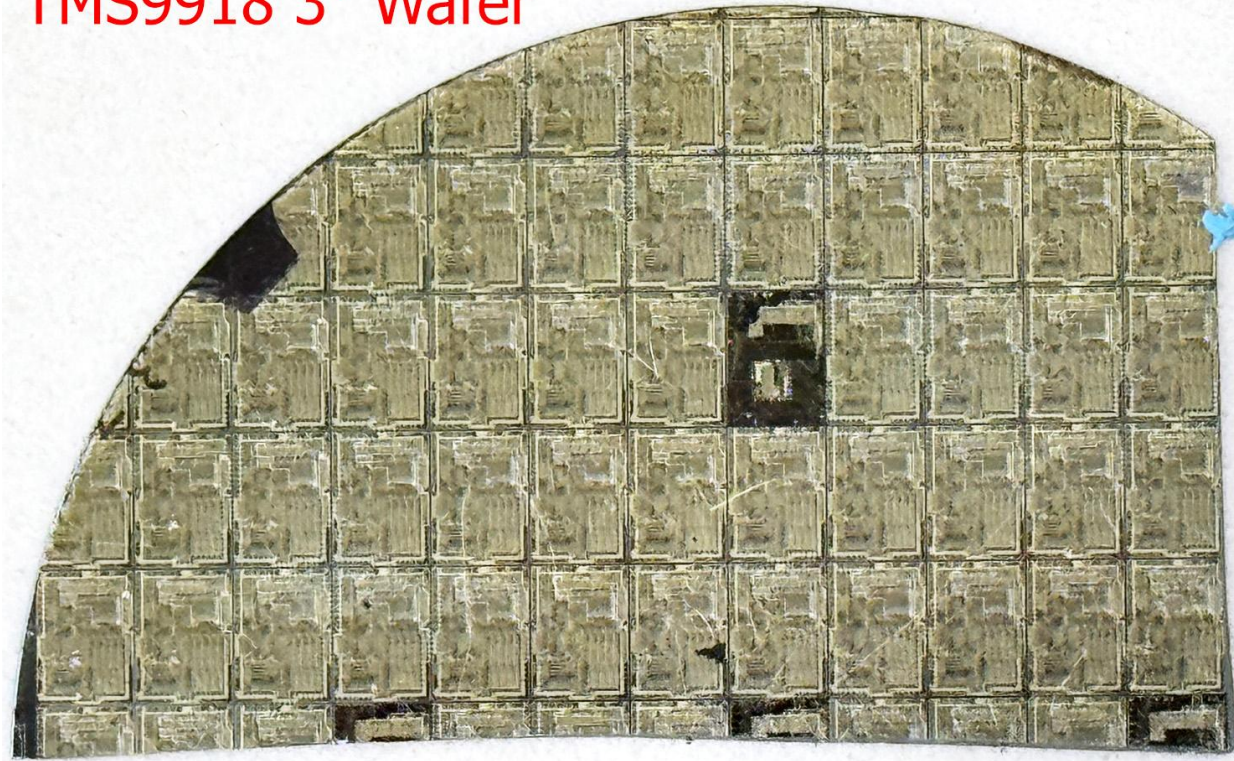


FIGURE 2-1 — VDP BLOCK DIAGRAM

9918 Die Photo
with a (rough)
overlay of
functional blocks



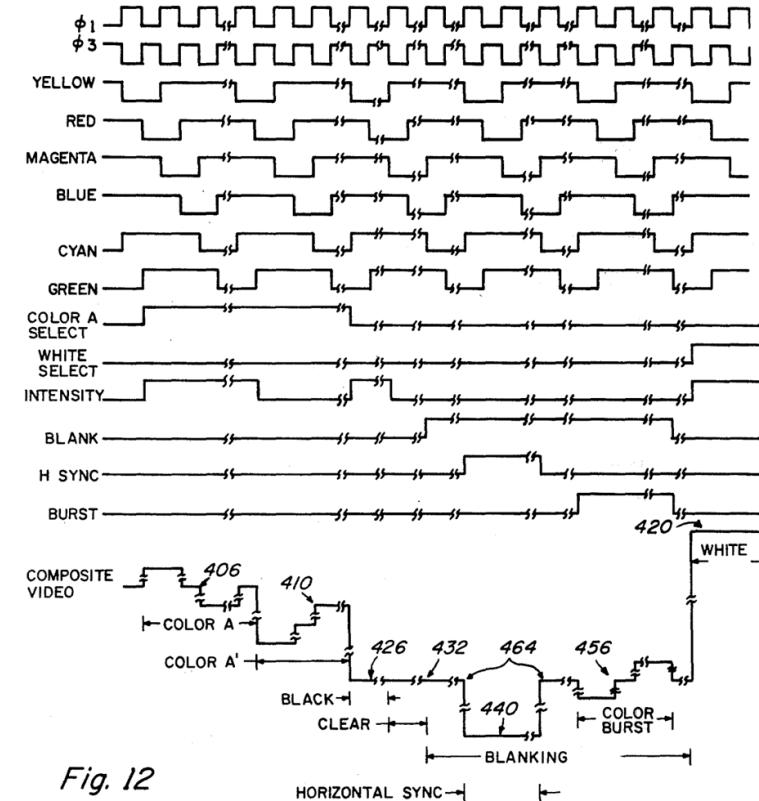
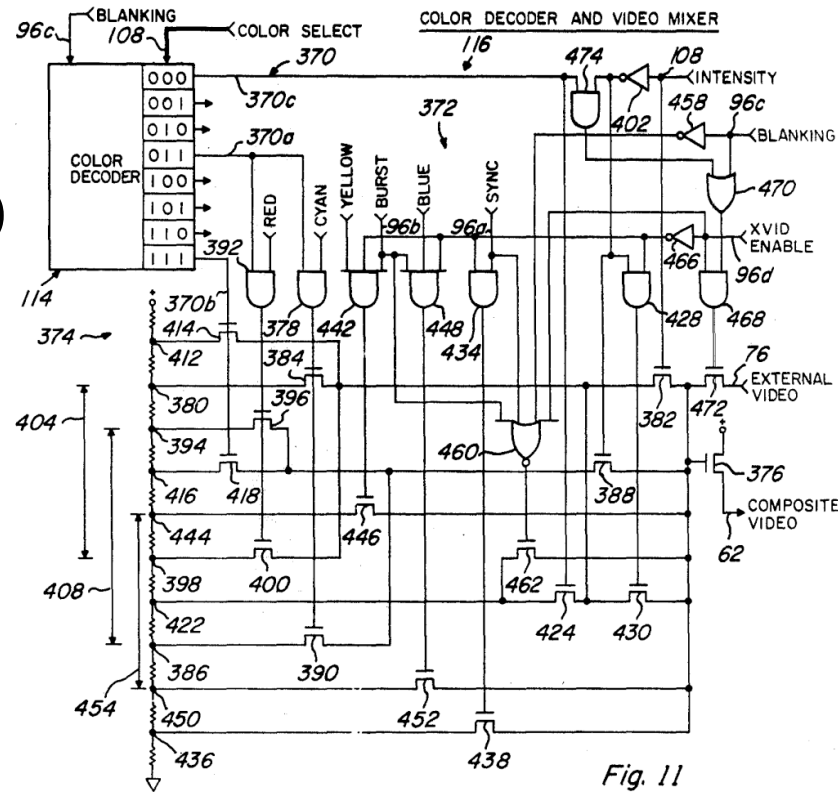
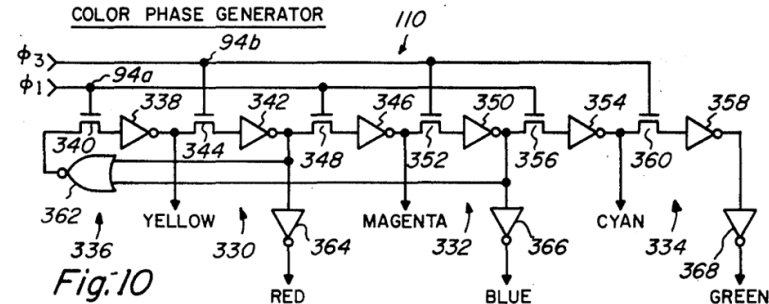
TMS9918 3" Wafer



9918 Video Circuitry

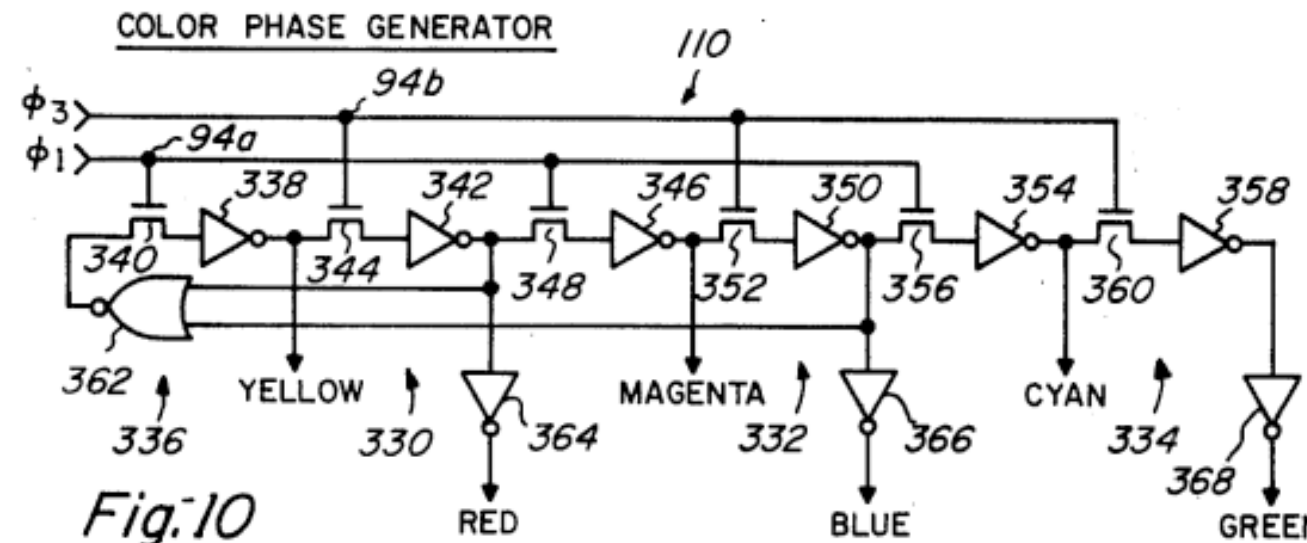
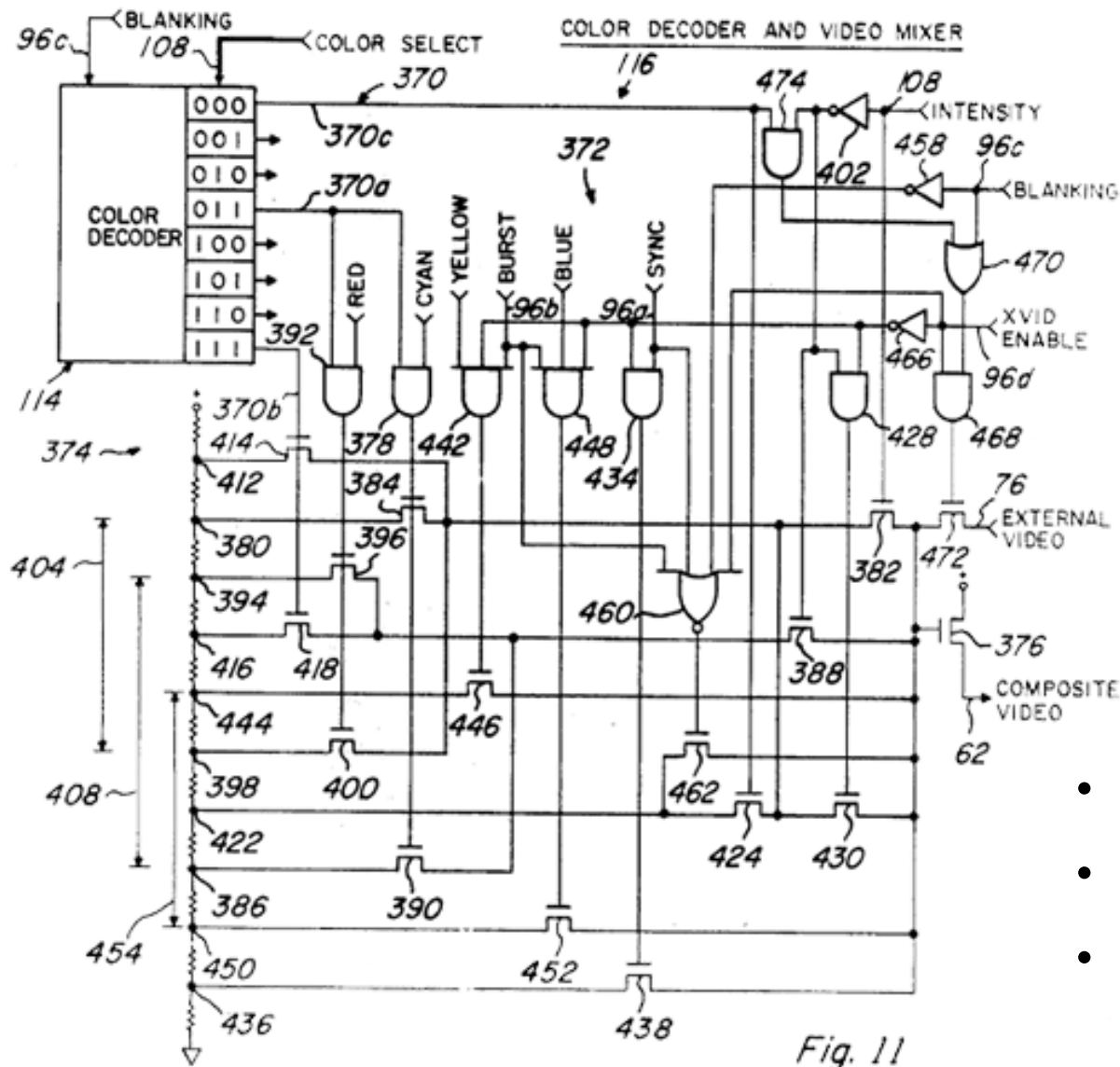
- Simple design
- Could tweak amplitude by tapping long resistor (simple voltage divider)
- Could tweak color phase by adjusting buffer strength/speed
 - Colors were tweaked based on feedback from home computer group

From VDP Patent 8,243,964 to Joe Sexton



The 9928 and 9929 worked similarly but there were three resistor trees to support the Y, R-Y, & B-Y outputs

Video Generation

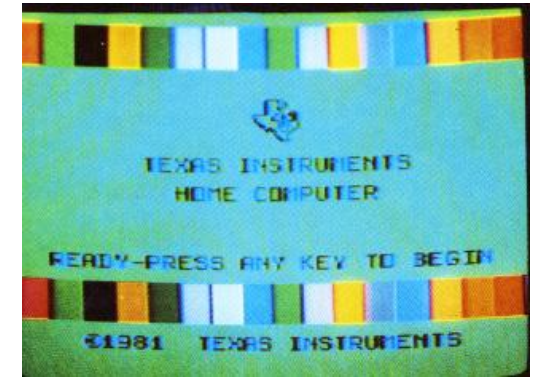
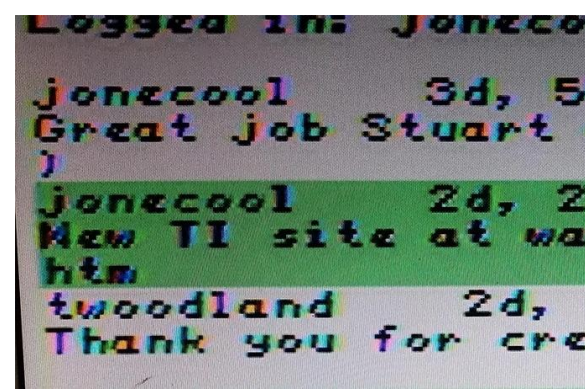


Patent 4,243,984

- Simple design
- Could tweak amplitude by tapping resistor
- Could tweak color by adjusting inverter buffers
 - Colors were tweaked based on feedback from HCG

9918 Video

- 256x192 pixels
- Pixel clock = color burst x 3/2
- As far as I know we got this scheme from the TI-HCG
- 262 TV scan lines (including blanking) Non-Interlaced
 - Hits same line twice with black between line
 - Causes "rainbow" due to not being $262\frac{1}{2}$ (525 interlaced) and thus color burst does not flip 180-degrees between frames
- Colors were tweaked as desired by TI-HCG
- 9928/29 used asynchronous external color burst to eliminate the "rainbow"
 - Coleco used the 9928 for this reason
- Using $\sim 5.4\text{M}$ pixel clock and 256 pixels got cut off by over scanning on many TVs cutting off the first letter of text – Thus the TI Home Computer Monitor



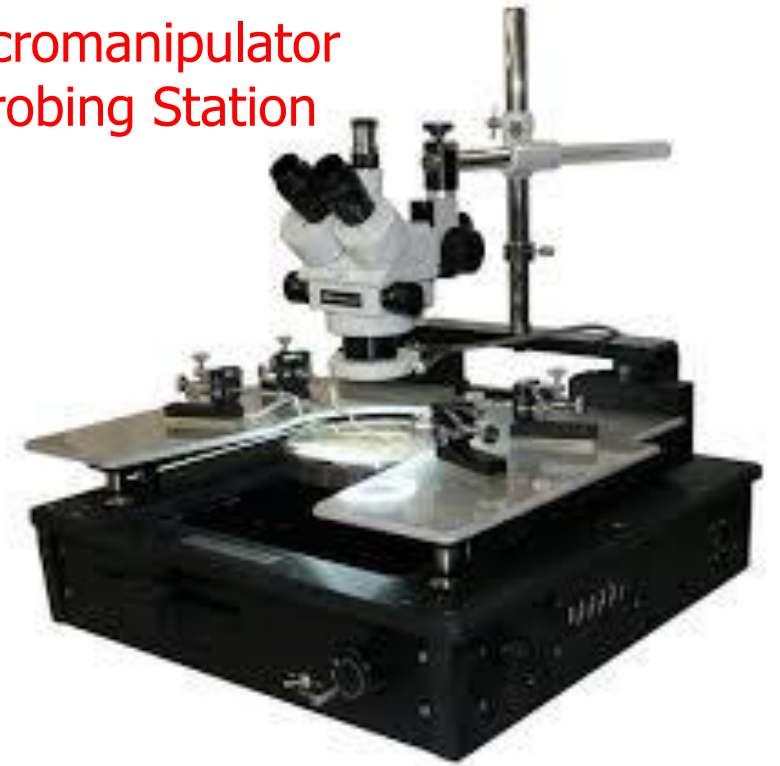
Things That Were Not Done

- Scrolling the background
 - We considered it but thought it might take too much hardware
- Full bitmap
 - 4-bit/pixel not possible due to memory and bandwidth
 - $256 \times 192 = 49,152$ pixels or 24,576 bytes
 - 9918A Graphics mode 2 (which I defined was a quick hack)
- 5 Sprites per line versus 32 per screen & 4 per line
 - Both were about equal in hardware

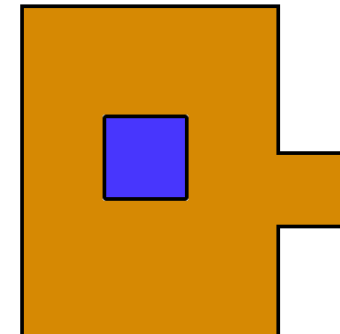
Debugging the TMS9918

- First Silicon
 - Would almost work by playing with the power supply voltage
- Found the device to work by inserting a probe near the horizontal timing logic (expected to be power hungry)
- Ship some packaged prototypes with stitch-bond and later a 2nd metal layer patch (perhaps a few hundred units)
- Found the stitch bond drawing current levels indicative of a resistive short
- By cutting various lines under a microscope, the high current area was isolated to a small region
- A non-simple figure was identified on the composite drawing that was causing a resistive short circuit
 - Likely a digitizing error
- Only one major mask needed to be fixed (depletion/diffusion)

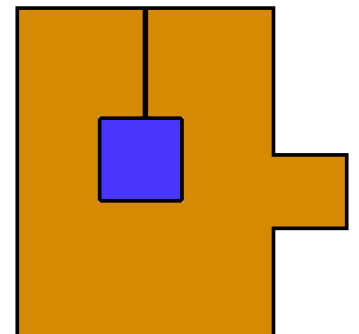
Micromanipulator
Probing Station



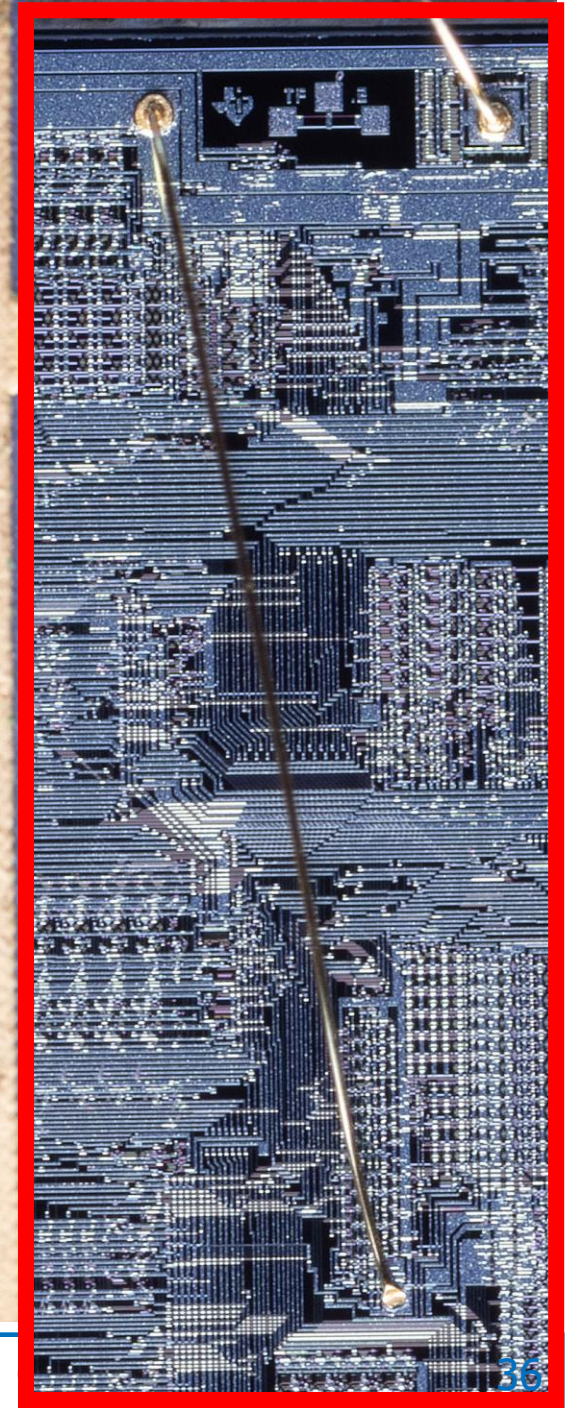
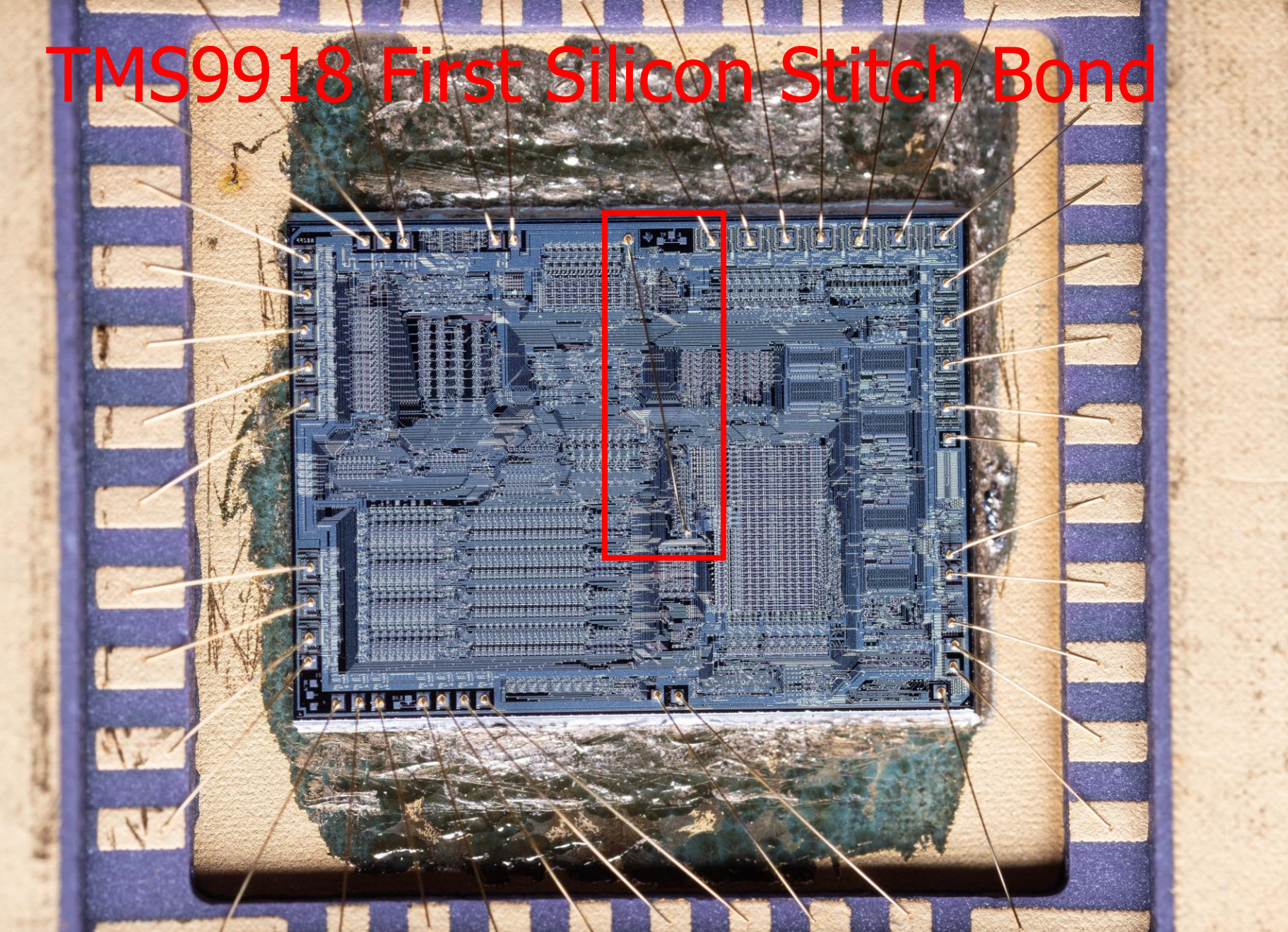
"Non-Simple Figure"



"Simple Figure"



TMS9918 First Silicon Stitch Bond



Graphics Mode 2

Super Patterns Graphics Mode

The proposed new mode for the VDP is more flexible and versatile than any of the existing modes, yet requires only very minor modification to the VDP. It gives more flexible display capability than the normal pattern mode with full resolution capability and gives better color capability than the multicolor mode. It gives a better display than the other 2 modes combined. This mode makes for very straight forward memory mapping of the display and may in fact give high enough color resolution to be straight forward enough so that computers could directly translate a TV camera's image for VDP displays. Note this capability can greatly reduce the amount of hard work necessary to generate a display image for any display and can eventually be a home computer option, since with the advent of video tape recorders color TV cameras will be in the home.

In general, the current display modes do exercise the full display capability, the way this new mode will. This new mode requires no modifications to any display ^{logic} and does not change any basic timings on the VBE. The Super Pattern mode merely makes minor modifications in how pattern addresses are formed and requires less than 50 transistors including the control logic, yet these minor modification have a great screen effect.

This new display mode is not exactly copyable by a low cost VDP with on board color table and fewer memory accesses because the new mode make use of the current VDPs memory cycles to access up to 6K of color table; however, the low cost VDP could subset the feature very easily by assuming a memory mapped display (the most likely way to use the new feature) thus the NAME_FETCH can be replaced by an on-chip color table fetch and the same net screen effect achieved. The two VDPs would be software compatible since the low cost VDP will be a subset of

The original VDPs capability; But not using the added feature of NAME mapping in the original VDP Software, the low cost VDP will work exactly as per the original VDP.

Description of Super Pattern Mode

In the Super pattern mode 3 consecutive NAME TABLES are used so that every 8x8 cell of the screen can have a different NAME. There ARE 768 cells on the screen (32 rows by 24 columns). In the current pattern mode there can only be 25 different shapes to fill the 768 cell, thus shapes must be used over an average of 3 times, the SUPER mode eliminates this restriction.

Description of the Super Pattern Mode

In the current pattern mode there are 768 cells (24 Row by 32 columns) on the screen, and yet only 256 NAMES are possible, thus shapes must be used over an average of three times. Also, the pattern mode restricts color options by having every 8 NAMES use the same 2 colors and each 8x8 cell on the screen can have only two colors. In some applications these color and shape restrictions can be covered and in general cause headaches in trying to generate complex displays. The Super Pattern Mode eliminates these restrictions. First the Super Mode Breaks the 768 Cells into 3 groups (top, middle and bottom at the same) and each group number (2 bits from the vertical line counter) is concatenated to the front of the NAME to give 768 different NAMES (one possible for each cell). Note that by ordering the NAME table the screen can now be memory mapped, which was not possible before. Additionally, in the Super mode, each line of a pattern descriptor can have 2 different colors (as opposed to 2 colors per 8 patterns currently) and the each 8x8 square can have all sixteen colors.

in it at the same time. Note the best the multi-color mode can produce is 4 different colors at an 8x8 region at a time and the Super mode gives dot by dot programmability. The net effect is shown below in an example of an 8x8 cell.

EXAMPLE OF Best Possible 8x8 Pixel display
(Hex Color Representation Given)

[illegible]

PATTERN MODE	MultiColor Mode	Super Mode
1) ONLY 2 COLORS POSSIBLE	1) Very low Resolution	1) Requires up to 16K Memory
2) Color Grouping Restrictive	2)	
3) Most Repeat Shapes an Average of 5 TIMES		

The only disadvantage of the Super mode is it requires up to 12 K of memory, but this is always the price paid for detailed color graphics and less memory can be used as the programmer's option. The Super mode allows for storing of the various tables used for display as this has some subset modes that use less memory. This shortcoming is controllable by the CPU by values loaded into the table base register. This feature enhancement requires no hardware additions since it happens unless hardware is added to stop it.

Competitive Comparison.

The UDP appears to have for superior object oriented graphics (sprites) and better high resolution with color capability, but the current UDP is available in few parts that cannot have a memory mapped ~~the~~ T.V. screen in a high resolution mode. Even for simpler parts such as the Motorola 6847 gives a two color high resolution ~~the~~ memory map mode. The super pattern graphics mode would give the UDP a very flexible and easy to use way to memory map the T.V. screen, and would ~~also~~ give the UDP the best ~~memory~~ full color memory mapping capability of any existing part. Since memory mapped graphics is the most straight forward way to display pictorial information, this would give the UDP another clear advantage in display capability; more simply put, any display the competitors part can great the UDP can do better with the super pattern graphics mode ~~the is for the same part~~

Memo by KG outlining
"Super Pattern Mode" for
9918A (Graphics mode 2)

TMS9918 VDP Sprite and Video Generation Patents

United States Patent [19] **4,243,984**
Ackley et al. [45] **Jan. 6, 1981**

Mostly about Sprites

[54] **VIDEO DISPLAY PROCESSOR** 4,116,444 9/1978 Mayer et al. 340/747 X
[75] Inventors: **David A. Ackley**, Stafford; **Gerald D. Rogers**, Sugarland; **Peter H. Macourek**, Needville; **Karl M. Guttag**; **Ki Suk Chang**, both of Houston, all of Tex. 4,119,955 10/1978 Nichols . 340/703
4,127,849 11/1978 Okor 340/703
4,129,858 12/1978 Hara . 340/703 X
4,155,095 5/1979 Kirschner 340/703 X
4,156,237 5/1979 Okada et al. 340/799 X

[73] Assignee: **Texas Instruments Incorporated**, Dallas, Tex. *Primary Examiner*—David L. Trafton
Attorney, Agent, or Firm—Andrew J. Dillon; Stephen S. Sadacca; Melvin Sharp

[21] Appl. No.: **18,540**

[22] Filed: **Mar. 8, 1979**

[51] Int. Cl.³ **G06F 3/153**

[52] U.S. Cl. **340/703; 340/725; 340/747; 340/799**

[58] Field of Search 340/703, 725, 747, 799

[56] **References Cited**
U.S. PATENT DOCUMENTS

3,891,792	6/1975	Kimura .	
3,918,039	11/1975	Clark .	
3,944,997	3/1976	Swallow	340/703 X
3,955,189	5/1976	Thomson	340/799 X
4,034,990	7/1977	Baer .	
4,081,797	3/1978	Olson .	
4,107,664	8/1978	Marino .	
4,107,665	8/1978	Mayer et al. .	

[57] **ABSTRACT**
A digital computing system is disclosed having a monolithic microprocessor, a versatile input/output subsystem, a multi-medium memory subsystem, and a video display subsystem with audio generating capability. In the memory subsystem, a balanced mix of dynamic RAM, P-channel MOS ROM and N-channel MOS ROM are included to minimize cost without sacrificing performance. In the video display subsystem, a monolithic video display processor performs all RAM access functions, in addition to composite video generation. The resultant composite video signal, which may include audio and external video information, can be applied directly to a video monitor or RF modulated for use by a television receiver.

8 Claims, 14 Drawing Figures

United States Patent [19] **4,262,302**
Sexton [45] **Apr. 14, 1981**

[54] **VIDEO DISPLAY PROCESSOR HAVING AN INTEGRAL COMPOSITE VIDEO GENERATOR** 4,116,444 9/1978 Mayer 340/745
4,119,955 10/1978 Nichols 340/706
4,129,858 12/1978 Hara 340/707
4,139,838 2/1979 Inose 340/703
4,155,095 5/1979 Kirschner 340/703

[75] Inventor: **Joe F. Sexton**, Houston, Tex.

[73] Assignee: **Texas Instruments Incorporated**

[21] Appl. No.: **17,865**

[22] Filed: **Mar. 5, 1979**

[51] Int. Cl.³ **H04N 9/12**

[52] U.S. Cl. **358/10; 350/30; 350/81**

[58] Field of Search 358/10, 81, 82, 22, 358/30; 340/701, 708, 726, 721, 745, 704

[56] **References Cited**
U.S. PATENT DOCUMENTS

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3,891,792	6/1975	Kimura	340/726
3,918,039	11/1975	Clark	340/730
4,034,990	7/1977	Baer	340/721
4,081,797	3/1978	Olson	340/706
4,093,960	6/1978	Estes	358/10
4,107,664	8/1978	Marino	340/731
4,107,665	8/1978	Mayer	340/706

[57] **ABSTRACT**
A digital computing system is disclosed having a monolithic microprocessor, a versatile input/output subsystem, a multi-medium capability. In the memory subsystem, a balanced mix of dynamic RAM, P-channel MOS ROM and N-channel MOS ROM are included to minimize cost without sacrificing performance. In the video display subsystem, a monolithic video display processor performs all RAM access functions, in addition to composite video generation. The resultant composite video signal, which may include audio and external video information, can be applied directly to a video monitor or RF modulated for use by a television receiver.

24 Claims, 14 Drawing Figures

Back then you had 1 year to file after public disclosure

MEMORANDUM

25 May 1978

TO: Gene McFarland

COPY: Sam Brown
John Hayn
Rex Naden

Bill Phares
Wally Rhines
Jerry Rogers

FROM: Dave Ackley

SUBJECT: VIDEO DISPLAY PROCESSOR COMPARISON

ATTACHMENTS: A) Video Display Processor Comparison
B) Electronics Article on AMI S680A
C) GI Data Sheet on AY8900

The attached comparison of the TI VDP and competition is based on the Electronics article and GI data sheet attached. The key advantages of our VDP is equal or better resolution, at least twice as many colors, more flexible and powerful dynamic/movable figures (sprites). The only features that the competition has an edge on us are: (a) AMI's multicolor mode has more resolution (256x192 or 128x92 vs. our 64x48) but at the expense of colors (2 or 4 of 8 vs. our 15), and (b) GI's dynamic/movable figures can have more per line (8 vs. our 4) but at the expense of total number (8 vs. our 32) and size (8x8 vs. our 8x8, 16x16 and the equivalent of 32x32 at half the resolution).

The AMI part could be more cost effective for simple CRT controller applications or cheap home computers because of its on chip default character set. It lacks our color and animation (sprite) capability. These capabilities should give us a significant advantage in the consumer market. The GI part is probably our closest competition but lacks the range of colors and requires an external composite color chip.

DA/mw
attachments

Dave

Dave Ackley

John 241
File comparison chart for future reference
Roger H
Data could be useful if we mkt mst 9918 next yr
Jon

VIDEO DISPLAY PROCESSOR COMPARISON

TI - TMS 9918

AMI - S68047

GI - AY 8900

Used in Mattel's Intellivision

BOARDER

COLORS

15

2

8

TEXT MODE

DISPLAY

40x24

40x24

No Text

CHARACTER SIZE

5x7

5x7

Mode

CHARACTER SET

INTERNAL

N/A

64

See

EXTERNAL

256

64

Background

COLORS

15

2

Below

PATTERN GRAPHICS MODE(S)

BACKGROUND

RESOLUTION

256x192

128x96

160x96

PATTERN ARRAY

32x24

32x24

20x12

PATTERN SIZE

8x8

4x4

8x8

PATTERN SET

256

32

64

PATTERN COLLORS

15

8

8

DYNAMIC/MOVABLE FIG.

HIGH RES

LOW RES

RESOLUTION

256x192

128x96

N/A

160x96

NUMBER/SCREEN

32

32

N/A

8

NUMBER/LINE

4

4

N/A

8

FIGURE SET

256/64

256/64

N/A

64

FIGURE SIZE

8x8/16x16

8x8/16x16

N/A

8x8

FIGURE COLORS

N/A

8

MULTICOLOR GRAPHICS MODE

HIGH RES

LOW RES

RESOLUTION

64x48

256x192

128x192

N/A

COLORS

15

2

4 of 8

N/A

MEMORY

INTERNAL

EXTERNAL

64 Characters

TEXT ONLY

2Kx8 { Expandable

1Kx10

256x8

GRAPHICS

4Kx8 { to 16Kx8

6Kx8

256x12

COMPOSITE VIDEO

On Chip

On Chip

External

NOTE:

Number of colors includes shades and gray levels

DA,03-370

What about VIC/Commodore?

MOS VIC 6560 (Used in VIC 20)

- VIC Designed in 1977 when to market in 1979 in the VIC 20
- 16 kB address space for screen, character and color memory (only 5 kB points to RAM on the VIC-20 without a hardware modification)
- 16 colors (the upper 8 can only be used in the global background and auxiliary colors)
- Two selectable character sizes (8×8 or 8×16 bits; the pixel width is 1 bit for "hires" characters and 2 bits for "multicolor" characters)
- maximum video resolution depends on the television system (176 × 184 is the standard for the VIC-20 firmware, although up to 248 × 232p/464i is possible on the NTSC machine and up to 256 × 280 is possible on the PAL machine[1])
- 4 channel sound system (3 square wave + "white" noise + global volume setting)
- on-chip DMA
- two 8-bit analog-to-digital converter
- light pen support
- No player/object graphics (Sprite-like)
- required fast and expensive SRAM

VIC II (introduced 1982 for the Commodore 64)

- 16 kB address space for screen, character and sprite memory
- 320 × 200 pixels video resolution (160 × 200 in multi-color mode)
- 40 × 25 characters text resolution
- Three character display modes and two bitmap modes
- 16 colors
- Concurrent handling of 8 Movable Object Blocks ("Sprite-like") per scanline, each of 24 (wide) × 21 pixels at 1 bit/pixel (12 × 21 multicolor = 2-bits/pixel pixels are 2x as wide)
 - Processor can get more Movable Objects via interrupt and changing pointers
- Raster interrupt (see details, below)
- Smooth scrolling
- Independent dynamic RAM refresh
- Bus mastering for a 6502-style system bus; CPU and VIC-II accessing the bus during alternating half-clock cycles (the VIC-II will halt the CPU when it needs extra cycles)

Mid 1978 to 1981 – Three Sad Years for 9918

- Home Computer group “vetoed” selling to Mattel (Intellivision) in 1978/9
 - They didn’t want to worry volume constrained due to allocation
 - Then Home Computer bombed leaving us with no consumer customers
- Program moved to product engineering for yield improvement
 - Minimal support in marketing
- Marketed to “Industrial Controls”
- Yield improvement by TI’s product engineering
- A few “believers” in Product Engineering and Marketing kept it going
 - I gave occasional help (while working on the 9995, 99000, 34010, and VRAM)
 - Pete Macourek had left TI (Founded Rose Electronics)

1980 Advertisement (aimed at "Industrial Controls")

TMS9918A. For color applications never before possible with a single chip.

TMS9918A. For graphics terminals. Video games. Arcade games. Home computers. Industrial process monitoring. Drafting, animation and education systems.

TMS9918A. For generating excitement in sixteen distinct colors.

This 40-pin plastic DIP generates a standard National Television Systems Committee (NTSC) composite video signal to drive a color or black/white monitor — you can tie it to an ordinary home TV antenna — simply by using a suitable RF modulator. The low-power TMS9918A requires only a 5-V supply.

Another leadership peripheral circuit in TI's 16-bit 9900 Family, the TMS9918A is designed with a general 8-bit data bus, allowing interface to virtually any CPU.

3-D capability

TMS9918A allows 3-dimensional simulation through thirty-five prioritized display planes. This unique feature

TMS9927 Video/Timer Controller

For high-resolution CRT terminal applications, TI offers the industry standard. A direct second source of SMC's CRT5027, the TMS9927 provides programmable timing and display for standard and non-standard CRT monitors, in both interlaced and non-interlaced formats. TMS9927 and TMS9918A — a powerful combination for full video capability — from Texas Instruments.

allows objects on the screen to pass in front of one another, just like they were on different planes.

An external video input pin lets you input a standard broadcast signal, have the TMS9918A overlay text/graphics, and output the resultant mix to a color display.

Imagine the possibilities this feature offers: subtitles, interactive broadcasting — and more.

Do-it-yourself flexibility

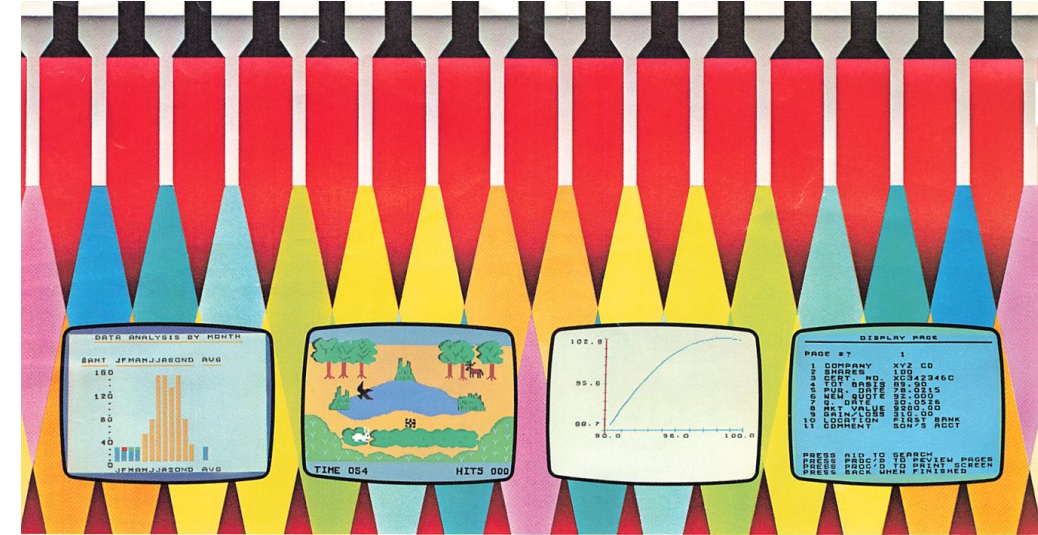
With the patterns defined in RAM, TMS9918A allows quick, easy alternate pattern set implementation. A powerful graphics mode allows complex graphics presentations, utilizing all 16 colors.

Thirty-two pattern objects can be moved smoothly across the screen to a positional resolution of one picture element. For textual applications, a flexible text mode provides resolution of 24 lines of 40 characters.

A typical video subsystem consists of a TMS9918A, eight dynamic RAMs and two TTL devices to generate CPU-9918A select signals.

The TMS9918A is available right now at your nearest authorized TI distributor.

For more information, write to Texas Instruments Incorporated, P.O. Box 1443, M/S 6404, Houston, Texas 77001.



Color it TMS9918A. New video display processor. From Texas Instruments.

TMS9918A. For color applications never before possible with a single chip.

TMS9918A. For graphics terminals. Video games. Arcade games. Home computers. Industrial process monitoring. Drafting, animation and education systems.

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85612

TO: Wally Rhines

COPY: Jim Huffhines
John Hughes
Ron Norris
Jerry Rogers

FROM: Karl Gutttag

SUBJECT: THE VIDEO DISPLAY PROCESSOR DEBACLE

I am making this report on what I consider to be the near disastrous state of affairs with the 9918 VDP and its associated spin-offs. We have taken a potential big money and image maker for MOS Microprocessor and turned it into another embarrassment. There has been an almost total lack of cooperation and communication between the design, products engineering and marketing areas with a lot of finger pointing at each other for who is responsible for the problems and almost nothing being done to correct them. I feel that there is still an opportunity to salvage some success from the VDP, if and only if corrective action is taken immediately.

First, as to the current state of the VDP; we have had fully functional VDP's for 2½ years, and we still are not in a position to ship production units. I am informed by Marketing that all major customers have disengaged with the VDP due to poor picture quality. Support for customers is a sad joke; generally, this amounts to an RTC person with an out-of-date spec., who is lucky to have seen a VDP picture, "helping" our customers. A customer must design his own system in order to investigate the VDP's capabilities; there are no application notes and little other written material to help the users. Our one chance to salvage the VDP in the U.S. appears to be the 9928A and it's being handled as a back door operation.

I now turn to what should be done. The 9928A is our only near term solution to the picture quality problems. With the 9928A, we should be able to eliminate the "rainbow effect" and be able to produce "zipperless" external video, the two major problems with picture quality. I am told that barring unforeseen delays or errors, we can have 9928A's in 8 weeks (P.G. is scheduled for about 4/20/81); I know that with proper expediting, this time can be brought in, but getting the 9928A is only half the job. We must have solid support for this part. While it would be easy to get a circuit to use the 9928A, we still don't have a hard and fast schematic or better yet, an evaluation board for the product. I consider the quick development of an evaluation board essential to success, since it lets the customer easily evaluate our product and will reduce customer questions; they will have a known working system rather than coming to us asking, "Will this work?" Incredibly, we do not have any of the original 9928's to work with in Houston to develop and/or test out any circuits. We must move on getting support for the 9928A now, which includes data sheets, evaluation boards and soliciting customers; as of now, we are standing around "flat footed" in all these areas.

THE VIDEO DISPLAY PROCESSOR DEBACLE
April 23, 1981
Page 2

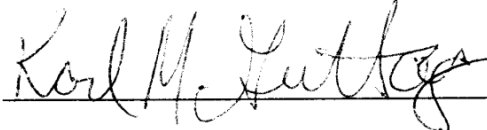
There needs to be a dynamic person assigned full time to making sure the right things happen and coordinating the activity; this person needs to have the authority to cut through the inter-manager bickering and get action started.

There are several additional issues that must be addressed, most importantly, the question of price elasticity vs. picture quality (i.e., is there a market for the 9918A and at what price?). Marketing appears to have no feel for this. My impression has always been that we have pushed the VDP toward higher end applications that are more quality rather than cost and feature conscious. We must solve yield problems with the 9918A and 9928A through proper levels of effort.

In conclusion, I must point out that the market window for the VDP is closing. While the new NEC part is not a direct competitor for the bulk of the VDP market (due to high system costs), I am relatively certain that Motorola and other companies will be introducing new display processors that will severely limit the VDP's market within the next year or so.

At a conference in which I presented the 9918 over a year ago, a marketing manager from Motorola came up to me and said, "It looks like a very nice part you have there. It's too bad you work for TI because they will never be able to sell it." So far, he has been proven right.

:hd


Karl Gutttag

9118/28/29 Support for By-4 DRAMs

MEMORANDUM
March 16, 1982

Request for By-4 DRAM Support
leading to 9118/28/29

TO: Ron Norris

COPY TO: Wally Rhines
John Hughes
Joe Yeager
Kevin McDonough

Granville Ott
Duke Hopkins
Pete Rickert
Jerry Rogers

FROM: Carlo Herrmann

SUBJECT: EXTENDING PRODUCT LIFE CYCLE OF 9918A/28A/29A VIDEO PRODUCTS

Wider word memories are on the horizon and with customer goal to minimize his product costs, it is important that we evaluate how to keep our video product line competitive longer!

Currently the 18A/28A/29A use a 7/7 row column address partitioning for 16K DRAM interface and 6/6 for 4K DRAMs. Two essential modifications will be necessary:

1. The addressing scheme needs to support the 4416 6/8 scheme as a mask programmable option. Presently the VDP uses 8 X TMS4116, a total of 9 chips. With this change the VDP will work with 2 X TMS4416, a total of 3 chips. Also with 8K X 8 chips it will be possible to have 2 chip minimum configuration.
2. The addressing reach of the VDP must be increased to 64K instead of 16K as a mask programmable option. With most games going to more complex screen generation schemes and the requirements for more software area the 16K memory size will be a limitation.

The above changes produce no pin configuration changes.

- Modification 1 requires that the address be shifted out differently by modifying the output multiplexer that feeds the ADO-AD7 lines.
- Modification 2 requires addition of hardware and interconnect to support the 8th bit.

I believe that we could either incorporate the changes to the current bar or initiate a cost reduction program, also important to compete more effectively.

Another enhancement that needs to be studied is the addition of a "bus available" signal. Presently the CPU is blind to when it can capture the memory address bus making it dependent on the very slow indirect-thru-the-VDP accessing scheme. This severely restricts how fast you can execute software and update the VDPs screen memory.

I urge that product engineering evaluate what it takes to incorporate both memory addressing and accessing changes in order to defend itself proudly in the 1980 video arena.

Carlo Herrmann
Carlo

CH:bdr

*Karl FYI, what do you think?
NOW inputs for cost red,
rainbow effect, and wide memory
but worth it?
KC*

1982-1983 Big Volume

- 1981 99/4A relaunch
- 1982 ColecoVision
- 1983 MSX Computer (ASCII-Microsoft)
- 1983 Sega 1000 & 3000
- Plus, many others



Texas Instruments
Home Computer
99/4 & 99/4A



Many Companies in Japan
Making MSX Computers



1983 Renewed Interest in a VDP

- ASCII-Microsoft (MSX) wanted more advanced VDP
 - Discussed feature sets with TI and several companies in Japan
 - Yamaha 9938 with 8 sprites/line and later the 9958
- TI developed 9228/AVDP (Advanced VDP) 1983-1985
 - ~8 Sprites/line, full 256x210 4-bit bitmap with 512 color palette, high-resolution table-driven mode (512 x 192 pixels) 80-character text mode
 - Ambition design with “risky design techniques”
 - Internal conflict and splitting resources with TMS34010
 - Managers at TI who had nothing to do with the 9918 until it was selling in selling in high volumes wanted the AVDP and didn’t understand the 34010
 - First silicon had many bugs that were never fixed and with Yamaha already delivering, the project was shelved

Legacy

- Nintendo NES Sprites were clearly influenced by 9918
 - Nintendo software on Colecovision
 - Sprite work very similarly
 - Sprite Pre-processing but with 8 per line
 - Sprites (only) 8 pixels wide 16 tall
 - Used **SRAM** rather than DRAM
 - NES introduced ~4 years after 9918



The TMS9918 and its variants were used in:
(includes Yamaha 9938 and 9958 Designs)

10 Amazing Texas Instruments TI-99/4A Facts

ColecoVision 9928

VTech CreatiVision

Memotech MTX

MSX / MSX 2 MSX=9918/28, MSX2=Yamaha 9938

Sega SG-1000 9918

Sega SC-3000 9918

Spectravideo SV-318 / SV-328

Sord M5

Tatung Einstein

Casio PV-1000 / PV-2000

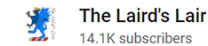
Tomy Tutor / Pyuta Jr.

DINA 2-in-1

Telegames Personal Arcade

Daewoo Zemmex

NABU Computer in Canada 9918



Subscribe

TMS34010 and "Bitmapped Sprites"

INTELLIGENT GRAPHIC CONTROLLER

- 0 THE IGC IS CONSIDERABLY DIFFERENT FROM ANY GRAPHICS CONTROLLER AVAILABLE OR ANNOUNCED TODAY.
- 0 THE IGC ARCHITECTURE IS SET UP FOR THE MANIPULATION OF VIDEO DATA STORED IN DYNAMIC RAMS. (EVEN WITH A VIDEO MEMORY THERE ARE STILL RELATIVELY FEW ACCESSSES AVAILABLE TO THE SCREEN MEMORY)
- 0 THE IGC HAS TWO MICROCONTROLLERS IN IT. THE FIRST CONTROLLER DOES THE FIELD EXTRACTION AND MANIPULATION OF THE SCREEN'S MEMORY DATA. THE SECOND CONTROLLER ACTS LIKE A GENERAL PURPOSE CPU TO MAKE DECISIONS ON WHERE AND WHAT DRAWING WILL OCCUR. THIS ARCHITECTURE WILL GIVE THE IGC BOTH HIGH PERFORMANCE AND PROGRAMMABILITY
- 0 THE IGC CAN EASILY EMULATE EXISTING GRAPHIC DEVICES SUCH AS THE 9918/28/29 VIDEO DISPLAY PROCESSOR.
- 0 THE IGC WILL OFFER THE GRAPHICS CAPABILITY THAT IS IMMEDIATELY NEEDED BY TERMINALS/WORKSTATIONS (TAM OF 4 TO 5 MIL. UNITS PER YEAR BY 1985). BY 1985 HOME TERMINALS/GAMES/TELETEXT WILL REQUIRE THE ADVANCED GRAPHICS OF THE IGC. (TAM OF 7+ MIL UNITS BY 1985)
(** NOTE MUST AIM AHEAD TO WHERE THE MARKET WILL BE IN 1985)

6/7/82 KMG 03. 910

Early TMS34010 (IGC) Presentation

TMS9918's Influence

- The word "Sprite" has become ubiquitous
- Clones and supersets of 9918
 - Sega and Nintendo
- Graphics Processors (TMS340 family)
 - TMS34010 started with "bitmapped sprites" (also known as bitblts/pixelblts)
 - Success of 9918/28/29 also worked against 34010 as "TI business people" wanted "more 9918"
- DRAM I/F on 9918
 - Multiport Video DRAM was influence by the bandwidth problems faced on the 9918
 - First it was simply display refresh
 - Interfacing the 9918 and 340xx to Analog DRAM cause me to push for a Synchronous DRAM inside TI
 - Video DRAM "shift register" implemented in SRAM led to the **Synchronous DRAM banks**
 - The SDRAM is essentially a M.P. VDRAM with the "serial port" coming out the parallel port pins
 - Processors had become so fast that display writing needed much more bandwidth than display refresh
- TMS9995 was design as the CPU for the 99/2 & 99/8
 - I moved from the 9918 to the 9995 (9995 designers designed TMS340xx)
 - The 9995's ALU was copied for use in the TMS32010 DSP (TI's First DSP)

99/4 CPU (MOS Design's Perspective)

- Home Computer Group (**HCG**) wanted their own custom CPU that would run GPL directly
 - Couldn't use another company's CPU (such as the Z80)
 - TI management vetoed a new instruction set
 - Home Computer kept GPL and would have the 99xx interpret it
 - Basic instructions were first interpreted into GPL before converting to 9900 assembly
- 9940 (8-bit data path running 16-bit 9900 instruction set) with EPROM
 - Plan was to spin off the 9985 by the EPROM with 256 bytes of SRAM
- 9940 design was a disaster taking the 9985 with it
- HCG had to choose between the 9900 and 9981 (byte wide external bus 9900 in 40-pins)
 - Chose 9900 for performance of running 16-bit wide to an external 256-byte SRAM
 - Essentially, the 9900 plus SRAM was working as a GPL interpreter
- **"Synergy" at TI could turn into the various groups hurting each other**

9900 Workspace Pointer

- Instruction set loosely Patterned after DEC PDP11
 - Workspace pointer moved registers to memory (saved cost, long term fatal flaw)
- Workspace Pointer point to register file in main memory
 - **A Reg to Reg Add takes 4 Memory cycles (Instr., Src-R, Dest-R, Dest-W)**
 - Touted as better for “context switches” (but only if the full register file needs to be swapped and the context switch is short)
- TI’s higher end 990 mini-computers cached the register file
 - Register File caching wasn’t feasible in a early 1980’s CPU
 - The Sun Sparc “RISC” (?) uses a similar concept with a cached “WP”
- Workspace pointer proved to be a burden
 - Forced too many memory cycles for non-WP-cached CPUs
 - Cache register file had too much overhead for a cost-effective single chip CPUs
 - Compiler technology moved in a different direction that leveraged register files

TMS9940 to TMS9985 Legacy

- TMS 9940 was a 16-bit microcomputer with embedded EPROM
 - 8-bit wide ALU and memory data path (*IMO saved few transistors but killed performance*)
 - Lead designer was a TTL designer and didn't leverage NMOS
 - Lots of business pressure and rushed P.G. releases
 - Many mask releases just to get to major opens or shorts - Pipeline of bugs
 - "There was no overlap in the process where both the CPU and EPROM would work"
 - There was no external data port – all test data was clocked in bit-serial using the CRU Instruction, which meant **the CPU had to work to test it and since the CPU had bugs it couldn't be tested**
 - Eventually the 9940CR, a complete redesign with a similar architecture was completed
- TMS9985 was spun out of the 9940 while there were still many bugs in the 9940
 - 9985 replaced the EPROM with 256 bytes of SRAM and had an external data path
 - 9985 was intended to be the CPU for the TI Home Computer
 - The 9995 started as the 9985A but was so different the name changed
- 9995 was 16-bit internally but with an 8-bit wide external data bus
 - Introduced instruction prefetch to the 9900 family
 - The 99/8 and 99/2 computers were to use the 9995 but never made it to market

9900 Family Clock Cycles										9900 Family Time Comparison						
						Instr., Src, & D On-Chip	9995 Instr. Off Src & D ON	Instr. & src off Dest On	Instr., SRC & Dest OFF	9900	9981	9985	9995			
	9900	9900	9981	9981	9985								Instr., Src, & D On-Chip	Instr. Off Src & D ON	Instr. & src off Dest On	Instr., SRC & Dest OFF
Clock Rate	333ns or 250ns	666ns or 500ns	400ns	800ns	500ns	333 ns	333 ns	333 ns	333 ns	0.666	0.800	0.500	0.333	0.333	0.333	0.333
Add	14	7	22	11	10	4	5	6	8	4.662	8.80	5.00	1.33	1.67	2.00	2.66
ABS MSB=0	12	6	16	8	12	3	4	6	6	3.996	6.40	6.00	1.00	1.33	2.00	2.00
ABS MSB=1	14	7	20	10	12	3	4	6	6	4.662	8.00	6.00	1.00	1.33	2.00	2.00
Increment	10	5	16	8	8	3	4	6	6	3.33	6.40	4.00	1.00	1.33	2.00	2.00
XOR	14	7	22	11	10	4	5	6	8	4.662	8.80	5.00	1.33	1.67	2.00	2.66
Clear	10	5	16	8	8	3	4	5	5	3.33	6.40	4.00	1.00	1.33	1.67	1.67
Invert	10	5	16	8	8	3	4	6	6	3.33	6.40	4.00	1.00	1.33	2.00	2.00
Multiply	52	26	62	31	82	23	25	26	29	17.316	24.80	41.00	7.66	8.33	8.66	9.66
Div (worst case)	128	64	136	68	126	33	34	35	39	42.624	54.40	63.00	10.99	11.32	11.66	12.99
Shift C≠0	2xC + 12	C + 6	2xC + 18	C + 9	2xC + 12	C + 7	C+8	C+9	C+9							
Jump PC Changed	10	5	12	6	6	3	4	4	4	3.33	4.80	3.00	1.00	1.33	1.33	1.33
Jump PC Not Ch.	8	4	16	8	6	3	4	4	4	2.664	6.40	3.00	1.00	1.33	1.33	1.33
Branch	8	4	12	6	8	3	4	4	4	2.664	4.80	4.00	1.00	1.33	1.33	1.33
Branch & Link	12	6	18	9	10	5	6	7	7	3.996	7.20	5.00	1.67	2.00	2.33	2.33
Br. & Load WP	26	13	38	19	20	11	12	14	17	8.658	15.20	10.00	3.66	4.00	4.66	5.66
9900 & 9981 - 5 micron NMOS, +5V, +12V, & -3V supplies										9995 and 9985 - 5 Micron NMOS, single +5V supply						

	9900					9995					9985			
	State	Memory	ALU	Other		State	Memory	ALU	Other		State	Memory	ALU	Other
	A1	Instruction		Instruction Decode		A1	Instruction				A1	Instr B1		
	A2	-	WP+2Souce			Prior I	Dest Write	WP+2S	Instruction Decode		A2	Instr B2	WP+2S B1	Instr Decode 1
	A3	Source				A2	Source	WS+2D			A3		WP+2S B2	Instr Decode 2
	A4	-	WP+2Dest			A3	Destination				A4	Src B1	WP+2D B1	
	A5	Destination				J1	Instruction	Add			A5	Src B2	WP+2D B2	
	A6	-	Add			A4	Dest Write				A6	Dest B1		
	A7	Dest. Write				J2		PC+Disp			A7	Dest B2	Add B1	
	J1	Instruction		Instruction Decode		Next I	Instruction				A8		Add B2	
	J2	-				J3			Instruction Decode		A9	Dest W1		
	J3	-									A10	Dest W2		
	J4	-	PC+Displacement								J1	Instr B1		
	J5	-									J2	Instr B2		
		Next Instr.									J3			
											J4			
											J5		PC+Disp1	
											J6		PC+Disp2	
												Next I		
Steps	12					7					16			

VRAM Memo

(Red Comment and Yellow Highlighting added 2025)

In 1982, TI had an internal messaging (email-like) system and most below the VP level share an MSG address where MSG were printed out on a common printer & hand delivered. I received this note on my email from Jerry Rogers.

A lot of MSGs back then were sent with ALL-CAPS

AMD = Advanced Memory Group, MPMC = Microprocessor & Microcontroller

-MSG M#= 81896 FR=MCD3 TO=JDR SENT=04/22/82 09:41 AM
R#=010 ST=C DIV=003 CC=0910 BY=MCD3 AT=04/22/82 09:41 AM

TO: BILL HOLTON ← OST
Strategic Planning Manager, TI Semiconductor

CC: KEVIN MCDONOUGH ← Became a VP at TI & President of Cyrix
WALLEY RHINES ← VP TI Semiconductor
DON BROOKS ← CEO Mentor Graphics '93
ROGER FISHER ← President of TSMC 1991-97
DICK GOSSEN AMD
DAVE LAFFITTE MPMC
DAN STENGER CSL

JERRY ROGERS ← Became a VP of TI
FOUNDER and CEO of Cyrix
Founded Silicon Display in 1988

JOHN HUGHES MPMC
LIONEL WHITE Senior TI Fellow
PEYTON COLE AMD
BOB WILLIAMSON MPMC
MARK NOVAK MPMC
BILL KNIGHT CSL

FM: KARL GUTTAG 4.5 years out of college, SMTS 1982, TI Fellow 1988, CTO of Silicon Display 1998

RE: WHY THE VIDEO MEMORY SHOULD BE ORGANIZED AS 64K BY 1
(RESPONSE TO YOUR MSG 91508)

MY INVESTIGATIONS INTO THE APPLICATIONS AND NEEDS FOR THE VIDEO MEMORY HAVE INDICATED THAT THE VIDEO MEMORY SHOULD BE ORGANIZED AS 64K BY 1. I WILL HIGHLIGHT HERE SOME OF THE REASONS.

THE MAIN MARKET FOR THE VIDEO MEMORY WILL BE WITH ROUGHLY 512 BY 512 DOT COLOR AND ROUGHLY 1024 BY 1024 DOT BW AND COLOR DISPLAYS. IT IS IN THESE APPLICATIONS WHERE THE HIGH BITS RATES GIVEN BY THE VIDEO MEMORY ARE MOST NEEDED (WHERE THE PART WILL HAVE ITS HIGHEST VALUE). THIS MARKET ALREADY HAS A SIGNIFICANT DEMAND FOR VIDEO MEMORIES AND IS EXPECTED TO GROW RAPIDLY IN THE 1980'S WITH PRODUCTS LIKE XEROX'S STAR WORKSTATION, THE IBM COLOR TERMINAL, DSG'S WORKSTATION, NEW TERMINALS FROM TPD, AND FUTURE PERSONAL COMPUTERS/WORKSTATIONS.

A RASTER SCANNED BIT MAPPED DISPLAY WITH 512 BY 512 RESOLUTION IN 16 COLORS (4 BITS PER COLOR) AND 1024 BY 1024 RESOLUTION IN BW (1 BIT PER COLOR) BOTH REQUIRE APPROX. 1 MILLION BITS OF DISPLAY MEMORY OR 16-64K MEMORY CHIPS REGARDLESS OF WHETHER THEY ARE BY 1 OR BY 4. GOING TO A BY 4 MEMORY WILL ADD MORE OUTPUT BUFFERS AND PINS THUS INCREASING THE COST OF THE DEVICE AND WILL MAKE FOR MORE COMPLICATED CONTROL CIRCUITRY FOR THE TARGET APPLICATIONS.

THE REASON WHY THE 4416 (16K BY 4) IS OF GREAT INTEREST TO CURRENT GRAPHICS MAKERS IS BECAUSE THEY CAN ACCESS ENOUGH BITS PER SECOND TO REFRESH THE DISPLAY AND STILL PROVIDE SOME ACCESSES FOR THE CPU'S OR GRAPHICS CONTROLLER'S UPDATE OF THE MEMORY. THEY GO TO A VERY WIDE BUS (64 BITS OR MORE) TO GET ENOUGH BITS PER SECOND AND BY USING A BY 4 MEMORIES THEY DO NOT HAVE TO BUY MORE MEMORY BITS THEN THEY NEED (IE BUY 64-64K BY 1 OR 64-16K BY 1 WHEN THEY ONLY NEED 16-16K BY 4 TO GET A 64 BIT DATA BUS) AND REDUCE THE PACKAGES PER SYSTEM. THE VIDEO MEMORY SOLVES THE "BITS PER SECOND" PROBLEM OF VIDEO DISPLAYS VIA THE HIGH SPEED SHIFT REGISTER AND AT THE SAME TIME GREATLY SIMPLIFIES THE SYSTEM DESIGN AND GREATLY IMPROVES THE NUMBER OF ACCESS THAT CAN BE GIVEN TO THE CPU; THE VIDEO MEMORY SOLVES SIGNIFICANT PROBLEMS FOR HIGHER RESOLUTION VIDEO APPLICATIONS THAT THE 16K BY 4 DOES NOT.

To: Karl Guttay
cc: Kevin M

RECEIVED
APR 22 1982
JERRY ROGERS

excellent report
Jerry

IGC was the early internal TI name for what became the TMS34010 GSP

THE MAIN MARKET THAT WOULD BENEFIT FROM A BY 4 VIDEO MEMORY WOULD BE THE LOW END GRAPHICS (GENERALLY HOME) SYSTEMS. A 256 BY 256 BIT MAP (ABOUT THE MAX RESOLUTION OF A COMPOSITE VIDEO SIGNAL) IN 16 COLORS REQUIRES 262K BITS OR 4-64K BIT MEMORIES. THUS A SYSTEM BUILT AROUND A 16 BIT GRAPHICS PROCESSOR (LIKE THE IGC) WITH 4-64K BY 4 VIDEO MEMORIES WOULD SEEM AT FIRST GLANCE IDEAL. SEVERAL FACTORS ARGUE AGAINST TARGETING THE VIDEO MEMORY AT THIS APPLICATION:

1. THE BITS PER SECOND RATE IS CONSIDERABLY LOWER SO A SLIGHTLY MORE COMPLICATED IGC COULD BE MADE TO WORK WITH 4416'S (ALTHOUGH WITH LOWER PERFORMANCE). IN THIS MANOR TI WOULD STILL BE COVERING LOW END SYSTEMS WITH THE 4416.
2. THE HOME SYSTEMS ARE CONSTANTLY MOVING UP IN FEATURES BY PROVIDING MORE COLORS AND BETTER RESOLUTION THUS REQUIRING MORE MEMORY. THIS WILL MAKE BY 1'S MORE ATTRACTIVE.
3. THE HOME SYSTEMS REQUIRE MORE RELATIVELY LOW COST MEMORY THAN JUST THAT USED FOR DISPLAY. BY GOING WITH 8 OR 16 BY 1 VIDEO MEMORIES THE EXTRA MEMORY SPACE CAN BE USED FOR OTHER STORAGE. A LOW COST SYSTEM CAN BE BUILT WHERE THE IGC ACTS BOTH AS THE GAME CONTROLLER (NOW DONE BY A MICROPROCESSOR) AND AS THE DISPLAY CONTROLLER WITH THE VIDEO MEMORY BEING THE ONLY RAM IN THE SYSTEM (THIS SOLUTION WILL BE LOWER COST THAN HAVING A G.P. MICROPROCESSOR WITH ITS MEMORY AND THEN A DISPLAY CONTROLLER WITH JUST ENOUGH MEMORY FOR DISPLAY ONLY).
4. WHILE THERE ARE MORE LOW END SYSTEMS SOLD PER YEAR (MAKING THIS THE LARGEST VOLUME MARKET FOR THE IGC), THE AMOUNT OF MEMORY PER SYSTEM IS SMALL. THE LARGEST VOLUME MARKET FOR THE VIDEO MEMORY WILL TEND TOWARD THE HIGHER END SYSTEMS THAT WILL USE A MINIMUM OF 16 VIDEO MEMORIES PER SYSTEM.

SUMMARY/CONCLUSIONS

1. THE BY 1 MEMORY CONFIGURATION IS BEST FOR THE HIGHER RESOLUTION DISPLAYS THAT HAVE THE CRITICAL NEED FOR AND WILL BE THE LARGEST MARKET FOR THE VIDEO MEMORY.
2. THE 4416 CAN COVER THE VERY LOW END OF THE VIDEO MARKET THAT NEEDS THE BY 4 MEMORIES. This didn't happen, the TMS34010 only worked with Dual Port VRAMs as the frame buffer
3. THE IGC NEEDS TO BE ABLE TO WORK WITH EITHER 4416 (FOR LOW END SYSTEMS) OR THE VIDEO MEMORY (FOR HIGHER END SYSTEMS).

I SHOULD ADD THAT WITH A 256K BIT VIDEO MEMORY, IT MAY BE DESIRABLE TO KEEP THE 64K MEMORY GRANULARITY AND GO TO A BY 4 MEMORY ORGANIZATION.

REGARDS,
KARL GUTTAG

Other Stories

- TMS340 Family, the first graphics processor
 - Beating Intel's 82786
 - 34020 almost designed into the color Mac
 - Meeting Warnock (Adobe) and Steve Jobs (NeXT)
 - Siggraph in the 1980s
 - First programmable processor to directly access DRAM
 - First graphic processor to use the multiport VRAM
- Multiport VRAM (DRAM with a shift register port)
 - History what led to the VRAM
 - My part in a joint strategy with the TMS34010
 - How the VRAM directly led to the SDRAM and GDRAM
- TMS320C80 MVP Early Image Processor
 - Four programmable DSPs and a RISC CPU on one chip
 - First chip interface to SDRAM and SVRAM
 - Cited in over a dozen patent lawsuits (as late as 2022)
 - Microunity and patent trolls
- TI Strategy Fallacies
 - Leader in Bipolar fell behind in MOS
 - Morris Chang (left TI to found TSMC)
 - Using calculators as a process driver
 - P-channel metal gate MOS
 - Led to "perfecting buggy whip" technology
 - Using DRAM as a process technology driver for NMOS
 - Led to making follower DRAMs
 - Lousy process for making CPUs
 - Forced to use TI's fabs that were uncompetitive
 - TMS340 competitors got better process at lower cost
 - TI did everything in-house
 - FABs
 - Design Tools (we had to wait for the internal design tools group to copy the functionality that other had)
 - Centralized purchasing decisions (PCs and software)
 - Sun SPARC (at TI)
 - Convince TI that they were far behind Intel in process
 - Said Intel's RAM bits many times smaller than TI's
 - Waste of effort on Bipolar output buffers
 - Led to Rich Templeton become CEO and Chairman of TI

Other Stories (Continued)

- Big Company versus Startups
- Liquid Crystal on Silicon
 - Competition with DLP, Laser Beam Scanning (**LBS**), and MicroLEDs
- Augmented Reality Displays and Optics
 - Exposure of Magic Leap's Deception
 - >\$3B dollars invested in smoke and mirrors
 - Microsoft's HoloLens 2
 - Stupidity of using laser beam scanning
 - Display devices for AR
 - LCOS, DLP, LBS, MicroLED, OLED

The End

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